

Exam: Biochemistry

Year: 2015

Dr. Doc Tutorial

[24/09/2026 9:22 AM] Imaanaa Harmee:

▽ Which one is the largest particulate of the cytoplasm?

- A. Lysosomes
- B. Mitochondria
- C. Golgi apparatus
- D. Endoplasmic reticulum

Correct Answer: B. Mitochondria

Detailed Explanation:

In eukaryotic animal cells, **mitochondria** are the largest discrete, membrane-bound organelles (or "particulates") suspended within the cytoplasm.

While the **nucleus** is the largest overall subcellular organelle, it is surrounded by the nuclear envelope and is anatomically distinct from the cytoplasm itself.

The **endoplasmic reticulum (ER)** is a larger continuous *membranous system*, but **mitochondria** (which typically measure 0.5 to 1 micrometer in width and can extend several micrometers in length) represent the largest distinct *particulate* structure found in cytoplasmic fractions. **Lysosomes** are significantly smaller single-membrane-bound digestive vesicles, typically measuring 0.1 to 1.2 micrometers.

[24/09/2026 9:22 AM] Imaanaa Harmee:

▽ Mitochondrial DNA is

- A. Circular double stranded
- B. Circular single stranded
- C. Linear double helix
- D. None of these

Correct Answer: A. Circular double stranded

Detailed Explanation:

Mitochondrial DNA (mtDNA) consists of **circular, double-stranded** DNA molecules.

This structure is located inside the mitochondrial matrix and undergoes replication independently of nuclear DNA. This circular, double-stranded structure closely resembles the genomic architecture of prokaryotes (bacteria), supporting the endosymbiotic theory of mitochondrial evolution.

[24/09/2026 9:22 AM] Imaanaa Harmee:

▽ Each of the following numbered processes are involved in signal transduction pathways:

- I. Response
- II. Amplification
- III. Transduction

IV. Reception.

Which of the following represents the sequence of events in a typical signal transduction pathway?

- A. IV, II, III, I
- B. IV, I, III, II
- C. I, II, III, IV
- D. IV, III, II, I

Correct Answer: D. IV, III, II, I (*Reception* → *Transduction* → *Amplification* → *Response*)

Detailed Explanation:

Cellular communication via chemical messengers follows a highly coordinated, unidirectional sequence of events:

Reception (IV): The signaling ligand (e.g., hormone or neurotransmitter) binds specifically to its extracellular receptor.

Transduction (III): The ligand-receptor interaction triggers a conformational change that converts the extracellular signal into an intracellular signal (often via G-proteins and second messengers).

Amplification (II): Intracellular enzyme cascades (such as phosphorylation cascades) exponentially multiply the signal at each sequential step.

Response (I): The cell executes a specific biological response, such as altering enzyme activity or gene expression.

[24/09/2026 9:22 AM] Imaanaa Harmee:

▽ In mammalian cells rRNA is produced mainly in the

- A. Endoplasmic reticulum
- B. Ribosome
- C. Nucleolus
- D. Nucleus

Correct Answer: C. Nucleolus

Detailed Explanation:

Ribosomal RNA (rRNA) is transcribed, processed, and assembled into ribosomal subunits within the **nucleolus**—a highly specialized substructure located inside the eukaryotic nucleus. Inside the nucleolus, RNA polymerase I transcribes rRNA genes (located in the nucleolar organizer regions) into a large 45S pre-rRNA precursor, which is subsequently cleaved and modified to yield the mature 18S, 28S, and 5.8S rRNA species. (The 5S rRNA is transcribed in the nucleoplasm by RNA polymerase III and subsequently migrates into the nucleolus for subunit assembly).

[24/09/2026 9:22 AM] Imaanaa Harmee:

The cell's structural unit that serves as microtubules organizing center and facilitates division during mitosis is:

- A. Centriole

- B. Ribosome
- C. Endoplasmic Reticulum
- D. All

Answer: A. Centriole. In eukaryotic animal cells, the microtubule network begins its assembly and orientation at the centriole (located within the centrosome), which acts as the cell's primary microtubule organizing center (MTOC). During mitosis, the centrioles duplicate and migrate to opposite poles of the cell to organize the mitotic spindle apparatus, which is responsible for physically separating chromosomes.

[24/09/2026 9:22 AM] Imaanaa Harmee:

Which of the following processes involve the combining of a message from one signaling molecule with that of another to either enhance or inhibit a cellular effect?

- A. Signal transduction
- B. Signal integration
- C. Signal amplification
- D. Signal reception

Answer: B. Signal integration. Signal integration (often involving "hormone cross-talk") is the process by which a cell receives multiple distinct extracellular signals and coordinates them into a single, unified cellular response. This allows the cell to reconcile conflicting or synergistic messages—either enhancing or inhibiting downstream metabolic pathways to meet the immediate physiological needs of the tissue.

[24/09/2026 9:22 AM] Imaanaa Harmee:

Nuclear DNA is present in combination with:

- A. Histones
- B. Non-histones
- C. Both (A) and (B)
- D. None of these

Answer: C. Both (A) and (B). Inside the eukaryotic nucleus, the long linear strands of double-stranded DNA do not exist in isolation. Instead, they are bound to a complex mixture of proteins to form chromatin. This nucleoprotein complex consists of highly basic histone proteins (H1, H2A, H2B, H3, and H4) that neutralize the DNA's negative charge and package it into nucleosomes, as well as a diverse array of non-histone chromosomal proteins (such as transcription factors, polymerases, and DNA repair enzymes) that regulate replication and gene expression.

[24/09/2026 9:22 AM] Imaanaa Harmee:

Extranuclear DNA is present in:

- A. Ribosomes
- B. Endoplasmic reticulum
- C. Lysosomes
- D. Mitochondria

Answer: D. Mitochondria. While the vast majority of cellular DNA is housed within chromosomes in the nucleus, eukaryotic cells also contain a small fraction of extranuclear DNA located in the matrix of the mitochondria. This mitochondrial DNA (mtDNA) is unique because it consists of closed, double-stranded circular molecules that replicate autonomously and encode specific proteins essential for oxidative phosphorylation.

[24/09/2026 9:22 AM] Imaanaa Harmee:

One of the following messenger molecules are derived from arachidonic acid?

- A. Corticoids
- B. Eicosanoids
- C. Steroids
- D. Terpenoids

Answer: B. Eicosanoids. Eicosanoids—which include prostaglandins, thromboxanes, and leukotrienes—are a class of highly potent, locally acting chemical messengers that reflect their origin from 20-carbon polyunsaturated fatty acids (specifically eicosa, meaning 20 in Greek). The primary precursor for these signaling molecules in humans is arachidonic acid, which is cleaved from membrane-bound phospholipids by phospholipase A₂ in response to tissue injury or inflammatory stimuli.

[24/09/2026 9:22 AM] Imaanaa Harmee:

Which one of the following is not under the category of cell surface receptors?

- A. Ligand-gated ion channels
- B. Enzyme-linked receptors
- C. G protein Coupled Receptors
- D. Intracellular Receptors

Answer: D. Intracellular Receptors. Receptors are divided into two main physical classes based on the chemical nature of their signaling ligands. Polar molecules (like peptide hormones, catecholamines, and cytokines) cannot easily cross the lipid bilayer and must bind to cell surface receptors embedded in the plasma membrane (such as ligand-gated ion channels, G-protein coupled receptors, and enzyme-linked receptors). In contrast, intracellular receptors reside in the cytoplasm or nucleus and are physically distinct from cell surface proteins.

[24/09/2026 9:22 AM] Imaanaa Harmee:

_____ is a type of receptor which responds to hydrophobic ligand molecules that are able to travel across the plasma membrane.

- A. Ligand-gated ion channel
- B. Enzyme linked receptor
- C. G protein Coupled Receptor
- D. Nuclear receptor

Answer: D. Nuclear receptor. Hydrophobic (lipophilic) signaling molecules, such as steroid hormones (cortisol, estrogen), thyroid hormone, and vitamin D, can readily diffuse across the hydrophobic plasma membrane. Once inside the cell, they bind to nuclear receptors (members of the steroid/thyroid hormone receptor superfamily).

These ligand-activated receptors then localize to the nucleus, where they bind directly to specific DNA hormone response elements (HREs) to modulate gene transcription.

[24/09/2026 9:22 AM] Imaanaa Harmee:

One of the following receptor is odd.

- A. Luteinizing hormone
- B. Calcitonin
- C. Calcitriol
- D. Nor-epinephrine

Correct Answer: C. Calcitriol

Biochemical Rationale:

Calcitriol (1,25-dihydroxycholecalciferol / active Vitamin D) binds to an **intracellular / nuclear receptor**. Because it is a lipophilic hormone, it diffuses through the plasma membrane and binds to a member of the steroid/thyroid hormone receptor superfamily of gene-specific transcription factors.

Luteinizing hormone, Calcitonin, and Norepinephrine all bind to **plasma membrane G protein-coupled receptors (GPCRs / heptahelical receptors)** on the cell surface

Ligand binding to G-protein coupled receptors

- A. activates phospholipase C- γ
- B. causes dissociation of G_{α} -GDP from the receptor followed by exchange of GTP for GDP
- C. stimulates G_{α} -GDP to exchange GDP for GTP, before dissociating from the receptor
- D. always causes elevation of intracellular $[Ca^{2+}]$

Correct Answer: C. stimulates G_{α} -GDP to exchange GDP for GTP, before dissociating from the receptor

Biochemical Rationale:

Mechanism of Activation: In the resting state, the inactive heterotrimeric G-protein ($G_{\alpha}\beta\gamma$) has GDP bound to its G_{α} subunit. When a ligand binds to the GPCR, the receptor undergoes a conformational change that acts as a guanine nucleotide exchange factor (GEF), prompting the G_{α} subunit to **release GDP and bind GTP**. Once GTP is bound, the active $G_{\alpha}\text{--GTP}$ complex **dissociates from both the receptor and the $G_{\beta}\gamma$ dimer** to interact with downstream effector enzymes like adenylyl cyclase.

Why the others are incorrect:

A is incorrect: GPCRs (G_q) activate phospholipase C- β (PLC_{β}), whereas phospholipase C- γ (PLC_{γ}) is activated by receptor tyrosine kinases like the insulin receptor.

B is incorrect: The exchange of GDP for GTP happens *before* the G_{α} subunit dissociates from the receptor.

D is incorrect: GPCRs linked to G_s or G_i regulate cAMP production via adenylyl cyclase, which does not always elevate intracellular calcium.

The G protein Coupled Receptors

- A. cytosine
- B. guanine
- C. Guanosine
- D. adenine

Correct Answer: C. Guanosine (or B. Guanine)

Biochemical Rationale:

Naming Origin: G-proteins are named "**G-proteins**" because they bind and hydrolyze **guanosine nucleotides**—specifically **guanosine triphosphate (GTP)** in their active state and **guanosine diphosphate (GDP)** in their inactive state. Guanosine is the purine nucleoside consisting of the nitrogenous base **guanine** linked to a ribose sugar.

One of the following protein receptor is activated by the binding of a ligand to a transmembrane receptor?

- A. Tyrosine kinase receptor
- B. Nuclear receptor
- C. G protein-coupled receptor
- D. Ligand-gated ion channel

Correct Answer: C. G protein-coupled receptor (or B. Nuclear receptor if asking which is NOT a transmembrane receptor)

Biochemical Rationale:

Plasma Membrane (Transmembrane) Receptors: Tyrosine kinase receptors, G protein-coupled receptors, and Ligand-gated ion channels are all plasma membrane proteins containing extracellular ligand-binding domains and membrane-spanning alpha-helices.

Nuclear Receptors: In contrast, Nuclear receptors are **intracellular receptors** located in the cytosol or nucleus; they do not span the plasma membrane. Lipophilic ligands diffuse across the lipid bilayer to reach them directly.

Note on question phrasing: If the question is asking which receptor transduces an extracellular signal via ligand binding to a 7-transmembrane helix to activate intracellular G-proteins, the answer is **C. G protein-coupled receptor**. If the question contains a common test-bank typo ("Which receptor is *NOT* a transmembrane receptor?"), the answer is **B. Nuclear receptor**.

Nuclear receptor

[24/09/2026 9:22 AM] Imaanaa Harmee:

Which type of receptor protein is activated by the binding of a ligand to an intracellular receptor?

- A. Tyrosine kinase receptor
- B. Nuclear receptor
- C. Ligand-gated ion channel
- D. G protein-coupled receptor

Correct Answer: B. Nuclear receptor

Biochemical Rationale: Lipophilic signaling molecules (such as steroid hormones, thyroid hormone, retinoic acid, and vitamin D) diffuse across the plasma membrane and bind to intracellular receptors. These receptors belong to the **nuclear receptor superfamily**, acting directly as gene-specific transcription factors that bind to hormone response elements (HREs) on nuclear DNA to regulate gene expression.

In mitochondria cristae act as sites for

- A. protein synthesis
- B. Oxidation-reduction reaction
- C. phosphorylation of flavoproteins
- D. breakdown of macromolecules

Correct Answer: B. Oxidation-reduction reaction

Biochemical Rationale: Cristae are convolutions of the inner mitochondrial membrane that greatly increase its surface area. They house the protein complexes of the electron transport chain (Complexes I–IV) and ATP synthase. As electrons flow down the chain toward oxygen, members of the chain undergo sequential **oxidation-reduction reactions** to establish a proton gradient for ATP synthesis.

Plasma membrane is composed of:

- A. Glycoproteins
- B. Lipids
- C. Chromoproteins
- D. Lipoproteins

Correct Answer: B. Lipids

Biochemical Rationale: The fundamental structural framework of all eukaryotic cell membranes is a fluid **lipid bilayer** composed predominantly of amphipathic phospholipids and cholesterol. Integral and peripheral proteins are embedded within or attached to this lipid matrix.

Which of the following is a short-lived messenger that acts by stimulating a soluble guanylyl cyclase raising [cGMP] and stimulating PKG?

- A. N₂O
- B. NO₂
- C. NO₃[–]
- D. NO

Correct Answer: D. NO

Biochemical Rationale: **Nitric oxide (NO)** is a lipophilic gas with a very short tissue half-life (3–10 seconds). It diffuses into vascular smooth muscle cells and activates cytosolic

soluble guanylyl cyclase, which converts GTP to cGMP. Elevated cGMP activates **protein kinase G (PKG)**, leading to vascular smooth muscle relaxation and vasodilation.

Anabolism and catabolism are chemically linked in the form of _____

- A. ATP
- B. ADP
- C. GTP
- D. Phosphodiester linkage

Correct Answer: A. ATP

Biochemical Rationale: Adenosine triphosphate (ATP) serves as the universal energy currency linking degradative and synthetic metabolism. **Catabolic pathways** break down fuel molecules to capture chemical energy in the form of ATP, whereas **anabolic pathways** consume the high-energy phosphate bonds of ATP to drive endergonic biosynthetic reactions.

[24/09/2026 9:22 AM] Imaanaa Harmee:

Which of the following statement true regarding gene Expression?

- A. The final product of gene expression is a protein
- B. Constitutive genes are not obliged to the environmental changes
- C. Operon is a segment of DNA in eukaryotic cells
- D. Effects have the same effect on gene expression

Correct Answer: B. Constitutive genes are not obliged to the environmental changes

Biochemical Rationale:

Constitutive genes (frequently referred to as "housekeeping" genes) encode products essential for basic, continuous cellular maintenance, meaning they are continually expressed regardless of environmental variations or physiological signals.

Why A is incorrect: The final product of gene expression can be either a functional noncoding **RNA** (such as rRNA, tRNA, or snRNA) or a **protein**, depending on the gene.

Why C is incorrect: **Operons are unique to prokaryotes** (bacteria); eukaryotic genomes do not contain operons, as each eukaryotic gene possesses its own dedicated promoter.

Gene Expression Regulation is important for the following changes except

- A. Cellular Differentiation
- B. Adaptation
- C. Morphogenesis
- D. Necrosis

Correct Answer: D. Necrosis

Biochemical Rationale:

Selective regulation of gene expression provides control over cellular structure and function, serving as the direct basis for **cellular differentiation**, **morphogenesis**, and cellular

adaptation to changing environments.

Necrosis is an accidental, unprogrammed form of cell death resulting from acute injury, toxins, or severe energy failure, rather than an active physiological process governed by gene expression regulation.

A segment of Operon that controls the overall processes of translation is

- A. Structural Gene
- B. Operator Gene
- C. Regulatory Gene
- D. P-Site

Correct Answer: B. Operator Gene

Biochemical Rationale:

In a bacterial operon, the **operator** is the cis-acting DNA segment that functions as the main regulatory switch.

When a repressor protein binds to the operator site, it physically obstructs RNA polymerase, preventing transcription of the polycistronic mRNA and thus halting downstream translation of all structural genes.

The Galactoside permease in Lac-operon is coded by

- A. The Y gene
- B. The Z gene
- C. The A gene
- D. The Y & Z gene

Correct Answer: A. The Y gene

Biochemical Rationale:

The *lac* operon contains three sequential structural genes:

lacZ gene: Encodes beta-galactosidase, which cleaves lactose into glucose and galactose.

lacY gene: Encodes **galactoside permease**, a membrane-bound transport protein that facilitates the movement of extracellular lactose into the bacterial cell.

lacA gene: Encodes **thiogalactoside transacetylase**.

Lactose is an inducer in lac-Operon; to exert its effect positively, it should bind with

- A. Repressor Gene
- B. Promoter site
- C. Repressor Protein
- D. Structural Genes

Correct Answer: C. Repressor Protein

Biochemical Rationale:

In the presence of lactose, a small amount is converted into its isomer, **allolactose**, which serves as the intracellular inducer.

Allolactose binds directly to the active **repressor protein** (the trans-acting product of the *lacI* gene). This binding induces a conformational shift in the repressor protein that prevents it from binding to the operator, allowing RNA polymerase to transcribe the operon.

[24/09/2026 9:22 AM] Imaanaa Harmee:

Which of the following compound is required for the attachment of RNA-polymerase to the P-site on lac-Operon?

- A. CAP
- B. CAP-cAMP
- C. Operator gene
- D. l-gene

Correct Answer: B. CAP-cAMP

Biochemical Rationale:

The lactose (*lac*) operon requires **positive regulation** for RNA polymerase to efficiently bind to its promoter (P-site).

When glucose is low or absent, intracellular cyclic AMP (cAMP) levels rise and bind to the **Catabolite Activator Protein (CAP)**. The resulting **cAMP–CAP complex** binds to the CAP-binding site upstream of the promoter, physically recruiting RNA polymerase and increasing transcription initiation efficiency.

Which of the following statement is not true regarding Iron metabolism regulation?

- A. mRNA for TfR has Cis-acting iron-responsive elements (IREs) at 3'-end
- B. IREs bound to trans-acting iron regulatory proteins
- C. When the concentration of iron in the cell is low, the IRPs bind to IREs
- D. When intracellular concentration of iron is high, IRPs bind to IREs

Correct Answer: D. When intracellular concentration of iron is high, IRPs bind to IREs

Biochemical Rationale:

Why D is FALSE: When intracellular iron levels are **high**, Iron Regulatory Proteins (IRPs) bind directly to iron rather than to Iron-Responsive Elements (IREs). Deprived of bound IRPs at its 3'-end, the mRNA for the transferrin receptor (TfR) becomes unstable and is rapidly degraded to prevent excess cellular iron uptake.

Statements A, B, and C are true: TfR mRNA contains 3'-cis-acting stem-loop IREs that bind trans-acting IRPs. Under **low iron** conditions, IRPs bind to IREs, stabilizing the TfR mRNA to allow transferrin receptor synthesis.

Which of the following amino acids would lead to sulfuric acid formation?

- A. Cysteine and isoleucine
- B. Cysteine and alanine

- C. Cysteine and methionine
- D. Methionine and isoleucine

Correct Answer: C. Cysteine and methionine

Biochemical Rationale:

Sulfuric acid (H_2SO_4) is a major nonvolatile inorganic acid generated during normal cellular metabolism.

It is produced specifically from the oxidative catabolism of the **sulfur-containing amino acids: cysteine and methionine**.

Which of the following groups of amino acids contain side-chain hydroxyl groups and could be a potential substrate for a protein kinase?

- A. Aspartate, glutamate, and serine
- B. Serine, threonine, and tyrosine
- C. Threonine, phenylalanine, and arginine
- D. Lysine, arginine, and proline

Correct Answer: B. Serine, threonine, and tyrosine

Biochemical Rationale:

Protein kinases regulate cellular pathways by catalyzing the transfer of a phosphate group from ATP to specific **hydroxyl ($-\text{OH}$) groups** on target proteins.

Serine, threonine, and tyrosine are the only three standard amino acids that possess polar side-chain hydroxyl groups capable of serving as phosphate acceptors.

All of the amino acids that are used to synthesize human proteins (with the exception of glycine) have which one of the following in common?

- A. An aromatic group
- B. A hydroxyl group
- C. An asymmetric carbon in the D-configuration
- D. An asymmetric carbon in the L-configuration

Correct Answer: D. An asymmetric carbon in the L-configuration

Biochemical Rationale:

Every standard amino acid incorporated into mammalian proteins contains an alpha-carbon bonded to four different chemical groups, making it a **chiral / asymmetric carbon atom**.

Human translational machinery exclusively uses amino acids in the **L-configuration**.

Glycine is the sole exception because its side chain is a simple hydrogen atom, meaning its alpha-carbon is attached to two identical hydrogens and lacks optical activity.

[24/09/2026 9:22 AM] Imaanaa Harmee:

In amyloidosis, α -helices may form alternative β -sheets of amyloid fibrils. The α -helices in these proteins can be characterized by which one of the following?

- A. They all have the same primary structure.
- B. They are formed principally by hydrogen bonds between a carbonyl oxygen atom in one peptide bond and the amide hydrogen from a different peptide bond.
- C. They are formed principally by hydrogen bonds between a carbonyl atom in one peptide bond and the hydrogen atoms on the side chain of another amino acid.
- D. They are formed by hydrogen bonding between two adjacent amino acids in the primary sequence.

Correct Answer: B. They are formed principally by hydrogen bonds between a carbonyl oxygen atom in one peptide bond and the amide hydrogen from a different peptide bond.

Biochemical Rationale: An **alpha-helix** is a regular secondary structural element stabilized by intrachain hydrogen bonds extending parallel to the helical spiral. These bonds form between the **carbonyl oxygen atom** of one peptide bond and the **amide hydrogen (–NH–)** of a peptide bond located four amino acid residues further down the chain. Because these hydrogen bonds form between atoms of the polypeptide backbone rather than variable side chains, alpha-helices can form from a wide diversity of primary sequences.

Which out of the following amino acids is a precursor for skin pigment?

- A. Tyrosine
- B. Histidine
- C. Phenyl Alanine
- D. Tryptophan

Correct Answer: A. Tyrosine

Biochemical Rationale: The protective dark pigment **melanin** is synthesized from **tyrosine** in the epidermis by pigment-forming cells called melanocytes. This pathway is initiated by the copper-containing enzyme **tyrosinase** (or tyrosine hydroxylase), which converts tyrosine toward DOPA and dopaquinone. Inherited defects in tyrosinase cause classic oculocutaneous albinism. While phenylalanine is the precursor to tyrosine, tyrosine is the direct precursor for melanin.

The following amino acid(s) is (are) glucogenic

- A. Leucine
- B. Lysine
- C. Alanine
- D. B and C

Correct Answer: C. Alanine

Biochemical Rationale: **Alanine** is a primary **glucogenic** amino acid that readily undergoes reversible transamination to yield **pyruvate**, a direct carbon substrate for gluconeogenesis. In contrast, **leucine** and **lysine** are the only two exclusively **ketogenic** amino acids; their catabolism yields acetoacetate, acetyl CoA, or acetoacetyl CoA, which cannot yield a net synthesis of glucose in human metabolism.

The protein present in hair is:

- A. Elastin
- B. Prolamine
- C. Gliadin
- D. Keratin

Correct Answer: D. Keratin

Biochemical Rationale: **Keratins** belong to a family of fibrous structural proteins whose conformation is predominantly alpha-**helical**. They serve as the major structural constituent of tough epidermal structures such as **hair**, skin, and nails, where their mechanical rigidity is determined by the density of interchain disulfide bonds between constituent cysteine residues.

Some amino acids are termed non-essential as they:

- A. Are not component of tissue proteins
- B. Have no role in metabolism
- C. May be synthesized in the body from 'essential' amino acids
- D. May be synthesized in body in prolonged fasting

Correct Answer: C. May be synthesized in the body from 'essential' amino acids

Biochemical Rationale: Amino acids are designated as **non-essential** (or nutritionally dispensable) because the body can synthesize sufficient quantities to meet physiological demands. They are synthesized either from common metabolic intermediates or directly from precursor **essential amino acids**. Classic examples include **tyrosine** (synthesized from the essential amino acid **phenylalanine**) and **cysteine** (which derives its sulfur atom from the essential amino acid **methionine**). Non-essential amino acids are active components of tissue proteins and participate extensively in cellular metabolism

[24/09/2026 9:22 AM] Imaanaa Harmee:

Which of the following is a dipeptide:

- A. Glutathione
- B. Oxytocin
- C. Homocysteine
- D. Aspartame

Correct Answer: D. Aspartame

Biochemical Rationale:

Aspartame is a synthetic **dipeptide** composed of aspartic acid and phenylalanine (specifically, L-aspartyl-L-phenylalanine methyl ester).

Glutathione is a **tri-peptide** (gamma-glutamyl-cysteinyl-glycine).

Oxytocin is a **nonapeptide** (a 9-amino-acid peptide hormone).

Homocysteine is an **amino acid** (a sulfur-containing non-protein amino acid intermediate in methionine metabolism).

Which of the following is not true about secondary protein structure?

- A. The hydrophilic/hydrophobic character of amino acid residues is important to secondary structure
- B. The ability of peptide bonds to form intramolecular hydrogen bonds is important to secondary structure
- C. The alpha helix, beta pleated sheet and beta turns are examples of protein secondary structure
- D. The steric influence of amino acid residues is important to secondary structure

Correct Answer: A. The hydrophilic/hydrophobic character of amino acid residues is important to secondary structure

Biochemical Rationale:

Secondary structure refers specifically to regular, recurring local folding patterns (such as the **alpha-helix**, **beta-pleated sheet**, and **beta-turns/bends**) stabilized by **intramolecular hydrogen bonds between atoms of the peptide backbone** (the carbonyl oxygen and amide nitrogen).

Steric constraints (bond rotation angles and side-chain hindrance) directly govern secondary structure formation.

In contrast, the **hydrophilic/hydrophobic character** (the hydrophobic effect) is the primary driving force for **tertiary structure** folding, where nonpolar side chains cluster into the deep interior core to avoid water while polar/charged side chains arrange on the aqueous exterior surface.

In non-competitive enzyme action:

- A. V_{max} is increased
- B. Apparent K_m is increased
- C. Apparent K_m is decreased
- D. Concentration of active enzyme molecule is reduced

Correct Answer: D. Concentration of active enzyme molecule is reduced

Biochemical Rationale:

In **non-competitive inhibition**, the inhibitor binds to a site distinct from the active site (binding either the free enzyme or the ES complex).

Because the inhibitor cannot be displaced by increasing substrate concentration, it functionally **lowers the concentration of active enzyme molecules** available to convert substrate into product.

Consequently, non-competitive inhibition **decreases the maximal velocity (V_{max})** while leaving the affinity for the substrate (K_m) **unchanged**.

[24/09/2026 9:22 AM] Imaanaa Harmee: **The type of reaction shown fits into which one of the following classifications?**

Correct Answer: A. Group transfer

Biochemical Rationale:

In this reaction, a glucosyl (glucose) residue is transferred directly from the activated nucleotide-sugar donor (**UDP-glucose**) to an alcohol group (R-OH) on an acceptor molecule, releasing **UDP**.

This fits the classification of a **group transfer reaction** because a functional group (a carbohydrate/glucosyl moiety) is transferred from one molecule to another.

Why other options are incorrect:

Isomerization (B) involves transferring atoms or functional groups within the *same* molecule. *Carbon-carbon bond breaking (C)* or *formation (D)* is catalyzed by lyases or ligases/synthases; here, the glycosidic bond cleavage and formation occur at an oxygen atom without breaking or forming C-C bonds.

The type of enzyme that catalyzes this reaction is which one of the following?

Correct Answer: C. Glycosyltransferase

Biochemical Rationale:

Enzymes that catalyze the transfer of an activated monosaccharide residue from a donor nucleotide sugar (such as UDP-glucose) to an acceptor molecule (such as a growing glycogen polymer, glycoprotein, or glycolipid) are termed **glycosyltransferases** (e.g., *UDP-glucose-glycogen glycosyltransferase*, commonly known as glycogen synthase).

Why other options are incorrect:

Kinase (A): Kinases specifically transfer high-energy **phosphate groups** (typically from ATP).

Dehydrogenase (B): Dehydrogenases transfer electrons as hydrogen atoms or hydride ions in **oxidation-reduction reactions**.

Isomerase (D): Isomerases rearrange atomic structure within a **single molecule** without net transfer between two separate substrates.

[24/09/2026 9:22 AM] Imaanaa Harmee:

Enzymes increase the rates of reaction by:

- A. Increasing the free energy of activation
- B. Decreasing the energy of activation
- C. Changing the equilibrium constant of the reaction
- D. Increasing the free energy change of the reaction
- E. Decreasing the free energy change of the reaction

Correct Answer: B. Decreasing the energy of activation

Biochemical Rationale:

Mechanism of Catalysis: Enzymes accelerate reaction rates by providing an alternate reaction pathway with a significantly lower **free energy of activation**. Lowering this energy barrier increases the proportion of substrate molecules with sufficient energy to pass through the high-energy transition state.

Why the other options are incorrect:

A is incorrect: Enzymes lower (decrease) the free energy of activation rather than increasing it.

C is incorrect: Enzymes accelerate the velocity with which equilibrium is reached, but they **do not alter the equilibrium constant** or the equilibrium ratio of products to reactants.

D & E are incorrect: Enzymes do not change the initial energy level of reactants or the final

energy level of products; therefore, the overall net free energy change of the reaction remains completely unchanged.

Lock-and-key” model of enzyme action proposed by Fisher implies that:

- A. The active site is flexible and adjusts to substrate
- B. The active site requires removal of PO₄ group

Correct Answer / Analysis: The active site is a rigid, pre-formed complementary template into which the substrate fits.

Biochemical Rationale:

Proposed by Emil Fischer, the **"Lock-and-Key" model** dictates that the active site contains amino acid residues arranged in a rigid, complementary three-dimensional surface that recognizes and binds the substrate, analogous to a key fitting into a rigid lock.

Option Analysis:

Option A ("The active site is flexible and adjusts to substrate") describes Daniel Koshland's **"Induced-Fit" model**, where substrate binding induces dynamic conformational changes in the flexible active site.

Option B is incorrect because catalysis does not universally require the removal of a phosphate group (which is specifically catalyzed by phosphatases).

The amount of adenine and uracil is equal to the amount of guanine and cytosine

- B. The amount of adenine and guanine is equal to the amount of uracil and cytosine
- C. The amount of adenine and guanine is equal to the amount of thymine and cytosine

[Chargaff's Rules Equivalence]

Correct Answer: C. The amount of adenine and guanine is equal to the amount of thymine and cytosine

Biochemical Rationale:

Chargaff's Rules for dsDNA: In double-stranded DNA, specific complementary base pairing dictates that the molar amount of **adenine equals thymine** ($A = T$) and **guanine equals cytosine** ($G = C$).

Combining these equivalences ($A + G = T + C$) shows that the **total amount of purines** ($A + G$) **equals the total amount of pyrimidines** ($T + C$).

Why options A and B are incorrect: Options containing uracil (U) apply to single-stranded RNA, which does not follow Chargaff's rules of equal base proportions.

[24/09/2026 9:22 AM] Imaanaa Harmee:

Which of the following statements is true?

- A. The template strand matches the sequence of the RNA transcript
- B. The two strands of DNA run parallel to each other
- C. G-C bonds are much more resistant to denaturation than A-T rich regions
- D. The common form of DNA is left-handed

Correct Answer: C. G-C bonds are much more resistant to denaturation than A-T rich regions.

Biochemical Rationale:

Thermal stability (melting temperature, T_m): Guanine–cytosine (G-C) base pairs are joined by three hydrogen bonds and possess stronger base-stacking interactions, whereas adenine–thymine (A-T) base pairs are joined by only two hydrogen bonds. As a result, DNA regions with high G-C content require significantly higher kinetic energy to separate into single strands (higher T_m) and are much more resistant to thermal denaturation than A-T rich regions.

Why the others are incorrect:

- **A is false:** The coding (non-template or sense) strand matches the sequence of the RNA transcript (with U replacing T). The template (antisense) strand is complementary to the RNA transcript.
- **B is false:** The two strands of the DNA double helix run antiparallel to each other (one 5'→3', the other 3'→5').
- **D is false:** The predominant, physiological form of DNA in human cells is B-DNA, which is a right-handed double helix. (Z-DNA is the rare left-handed form.)

A patient was diagnosed with a sphingolipidosis, which are congenital diseases involving the inability to degrade sphingolipids. All sphingolipids have in common:

- A. a glycerol backbone
- B. ceramide
- C. phosphorylcholine
- D. N-acetylneuraminic acid

Correct Answer: B. ceramide

Biochemical Rationale:

All sphingolipids are built upon the amino alcohol **sphingosine**. When a long-chain fatty acid is attached via an amide linkage to the amino group of sphingosine, it forms **ceramide**. Ceramide serves as the core structural precursor common to all sphingolipids, including sphingomyelin and complex glycosphingolipids (cerebrosides, sulfatides, globosides, and gangliosides).

Why the others are incorrect:

A (glycerol backbone): Glycerol forms the backbone of triacylglycerols and glycerophospholipids, whereas sphingolipids contain a sphingosine backbone.

C (phosphorylcholine): Phosphorylcholine is present in sphingomyelin, but not in glycosphingolipids.

D (N-acetylneuraminic acid / NANA): NANA (sialic acid) is specific to gangliosides.

A patient was diagnosed with a hypertriglyceridemia. This condition is named for the high blood levels of lipids composed of:

- A. fatty acyl groups attached to a glycerol backbone
- B. A glycerol lipid containing a phosphorylcholine group

- C. A sphingolipid containing three fatty acyl groups
- D. Three glycerol moieties attached to a fatty acid

Correct Answer: A. fatty acyl groups attached to a glycerol backbone

Biochemical Rationale:

Triacylglycerols (triglycerides) consist of **three fatty acyl chains esterified to a single three-carbon glycerol backbone.**

Hypertriglyceridemia refers to elevated circulating blood levels of these triacylglycerol-rich plasma lipoproteins (such as chylomicrons and VLDL).

A patient was diagnosed with a deficiency of the lysosomal enzyme glycosidase. The name of the deficient enzyme suggests that it hydrolyzes a glycosidic bond, which is a bond formed:

- A. Through multiple hydrogen bonds between two sugar molecules
- B. Between the anomeric carbon of a sugar and an O-H (or N) of another molecule
- C. Between two anomeric carbons in polysaccharides
- D. Between the carbon containing the aldol or keto group and the α -carbon

[24/09/2026 9:27 AM] Imaanaa Harmee:

Correct Answer: B. Between the anomeric carbon of a sugar and an O-H (or N) of another molecule

Biochemical Rationale:

A **glycosidic bond** is a covalent linkage formed between the **anomeric carbon** of a monosaccharide (in cyclic hemiacetal or hemiketal form) and either a hydroxyl group (**O-glycosidic bond**) or an amine nitrogen (**N-glycosidic bond**) of another compound.

Lysosomal glycosidases (acid hydrolases) cleave these specific bonds during the degradation of complex carbohydrates, glycosaminoglycans, and glycoproteins.

If the DNA strand shown below serves as a template for the synthesis of RNA, which of the following choices gives the sequence and direction of the RNA?

5'-GCTATGCACTGGTAGCAATTGCGT-3'

- A. 3'-CGUAUCGUAGCACUAGCUAAACGCA-5'
- B. 5'-UGGUUAAGCUUGAUGCUACGUUCC-3'
- C. 5'-ACGCAUUUCGAUACGAUGCUAAACGCA-3'
- D. 5'-CGUAUCGUAGCACUAGCUAAACGCA-3'

Correct Answer: A. 3'-CGUAUCGUAGCACUAGCUAAACGCA-5' (or C. 5'-ACGCAAUUCGAUCACGAUGCAUAGC-3' in standard 5'→3' notation)

Biochemical Rationale:

Antiparallel Transcription: RNA polymerase reads the DNA template strand in the 3'→5' direction to synthesize an RNA transcript in the 5'→3' direction.

If the template is given as **5'-GCTATGCACTGGTAGCAATTGCGT-3'**, its complementary RNA sequence generated antiparallel (from the template's 3'-end to 5'-end) is: **5'-ACGCAAUUCGAUCACGAUGCAUAGC-3'**

Aligning antiparallel directly opposite the 5'-end of the template yields **3'-CGUAUCGUAGCACUAGCUAAACGCA-5'** (Choice A).

Which of the following phospholipids act as surfactants in human lung:

- A. Dipalmitoyl lecithin
- B. Cardiolipin
- C. Phosphatidyl inositol
- D. Phosphatidyl serine

Correct Answer: A. Dipalmitoyl lecithin

Biochemical Rationale:

Dipalmitoyl lecithin (also known as **dipalmitoylphosphatidylcholine, DPPC**) is the major lipid component of pulmonary surfactant secreted by Type II pneumocytes.

It reduces surface tension at the alveolar air-liquid interface to prevent alveolar collapse (atelectasis) during expiration. A deficiency of DPPC in premature infants leads to **Infantile Respiratory Distress Syndrome (RDS)**.

Which of the following glycerophospholipid play important role in cell signaling?

- A. Cardiolipin
- B. Phosphatidic acid
- C. Phosphatidyl Choline
- D. Phosphatidylinositol

Correct Answer: D. Phosphatidylinositol

Biochemical Rationale:

Phosphatidylinositol 4,5-bisphosphate (PIP₂) is a membrane glycerophospholipid that plays a pivotal role in intracellular signal transduction.

Cleavage of PIP₂ by **Phospholipase C (PLC)** in response to hormone binding yields two second messengers: **Inositol 1,4,5-trisphosphate (IP₃)**, which mobilizes Ca²⁺ from the endoplasmic reticulum, and **Diacylglycerol (DAG)**, which activates Protein Kinase C (PKC).

Phosphorylation of PIP₂ by PI 3-kinase also generates PIP₃ in insulin and cell survival signaling pathways.

[24/09/2026 9:27 AM] Imaanaa Harmee:

Which of the following is not a component of glycosphingolipids?

- A. Carbohydrate
- B. Sphingosine
- C. Glycerol
- D. Fatty acids

Correct Answer: C. Glycerol

Biochemical Rationale:

Glycosphingolipids (such as cerebrosides, globosides, and gangliosides) are built upon the amino alcohol **sphingosine**. When a long-chain fatty acid is attached via an amide bond to

sphingosine, it forms **ceramide**. Attaching one or more sugar residues (carbohydrates) to the terminal hydroxyl of ceramide produces a glycosphingolipid.

Glycerol is absent in sphingolipids; it serves as the three-carbon backbone exclusively for **triacylglycerols** and **glycerophospholipids**.

Which of the following is not essential fatty acid?

- A. Oleic acid
- B. Linolenic acid
- C. Linoleic acid
- D. Arachidonic acid

Correct Answer: A. Oleic acid

Biochemical Rationale:

Oleic acid (18:1, Δ^9 , ω -9) is a monounsaturated fatty acid that human tissues can synthesize de novo from stearic acid using the microsomal Δ^9 -desaturase enzyme.

Linoleic acid (18:2, ω -6) and α -linolenic acid (18:3, ω -3) are essential fatty acids because mammalian cells lack the desaturase enzymes required to introduce double bonds beyond carbon 9 toward the ω (methyl) end. Arachidonic acid is synthesized from dietary linoleic acid and is considered conditionally essential.

Why the others are incorrect:

B is false: Linoleic acid (18:2, ω -6) is an essential fatty acid. Mammalian cells cannot introduce double bonds beyond carbon 9 toward the ω end, so it must be obtained from the diet.

C is false: α -Linolenic acid (18:3, ω -3) is an essential fatty acid for the same reason—humans lack the desaturase enzymes needed to synthesize it and must acquire it from dietary sources.

D is false: Arachidonic acid (20:4, ω -6) is conditionally essential. It can be synthesized endogenously from dietary linoleic acid, so it is not strictly required in the diet under normal physiological conditions.

Which of the following lipid is used for energy storage?

- A. Glycolipid
- B. Glycerophospholipid
- C. Sphingolipids
- D. Triacylglycerol

Correct Answer: D. Triacylglycerol

Biochemical Rationale:

Triacylglycerols (TAGs / triglycerides) consist of three fatty acyl chains esterified to a single glycerol backbone. Because they are completely uncharged and hydrophobic, they pack densely without water in the lipid droplets of adipose tissue, serving as the body's primary long-term metabolic energy reservoir.

Glycolipids, glycerophospholipids, and sphingolipids are structural components of biological membranes rather than fuel storage molecules.

Phospholipids are important cell membrane components because:

- A. They have both polar and nonpolar portions
- B. They have glycerol
- C. They can form bilayers in water
- D. They combine covalently with proteins

Correct Answer: A. They have both polar and nonpolar portions (or C. They can form bilayers in water)

Biochemical Rationale:

Phospholipids are **amphipathic** molecules possessing a hydrophilic (polar/charged) head group and long hydrophobic (nonpolar) fatty acyl tails.

This dual chemical nature causes them to spontaneously assemble into continuous **lipid bilayers** in aqueous environments, orienting their hydrophobic tails inward away from water and their polar heads outward toward the aqueous cytosol and extracellular fluid.

[24/09/2026 9:27 AM] Imaanaa Harmee:

The repeating unit in hyaluronic acid is:

- A. Glycerol and galactose
- B. Glycerol and N-acetyl galactosamine
- C. Glycerol and glucosamine
- D. Glycerol and N-acetyl glucosamine

Correct Answer: D. Glycerol and N-acetyl glucosamine

(Biochemical composition: D-Glucuronic acid + N-acetylglucosamine)

Biochemical Rationale:

Hyaluronic acid (hyaluronate) is an unbranched glycosaminoglycan (GAG) composed of repeating disaccharide units of D-glucuronic acid and N-acetylglucosamine (GlcNAc) linked by alternating $\beta(1\rightarrow3)$ and $\beta(1\rightarrow4)$ glycosidic bonds.

(Note: In this standard test-bank question, "Glycerol" in option D represents a common typographical error for "Glucuronic acid," but Choice D correctly identifies N-acetylglucosamine as the amino sugar constituent.)

Why the others are incorrect:

A is false: This option does not correctly identify the repeating disaccharide units of hyaluronic acid, which require both D-glucuronic acid and N-acetylglucosamine.

B is false: This option does not match the known composition of hyaluronic acid, which consists of D-glucuronic acid and N-acetylglucosamine.

C is false: This option fails to identify N-acetylglucosamine as the amino sugar component of hyaluronic acid.

Glucose and galactose are epimers and they differ structurally in orientation of H and OH on:

- A. C4
- B. C3

- C. C2
- D. C5

Correct Answer: A. C4

Biochemical Rationale:

Epimers are carbohydrate stereoisomers that differ in the spatial configuration of the hydroxyl (–OH) group around only **one specific asymmetric carbon atom**.

D-Glucose and **D-Galactose** differ strictly in the orientation of the –OH group at **carbon 4 (C-4)**, making them C-4 epimers. (In contrast, D-glucose and D-mannose are C-2 epimers).

The reducing ability of a carbohydrate is due to the presence/or formation of:

- A. A free carboxyl group
- B. A free hydroxyl groups
- C. Presence of a 1-4 linkage
- D. Formation of ring structure

Correct Answer: B. A free hydroxyl group (*specifically the anomeric hydroxyl group / free aldehyde formation upon ring opening*)

Biochemical Rationale:

A sugar acts as a **reducing sugar** if it possesses an unlinked, free **anomeric carbon atom** carrying a free hydroxyl group (in cyclic hemiacetal or hemiketal form).

In solution, mutarotation allows the sugar ring to open into a straight-chain acyclic form containing a free **aldehyde (or alpha-hydroxyketone) group**, which readily donates electrons to reduce mild oxidizing agents (such as Benedict's or Fehling's reagents).

Phosphatidic acid on hydrolysis yields:

- A. Glycerol, fatty acids, phosphoric acid, choline
- B. Glycerol, fatty acids, phosphoric acid
- C. Glycerol, fatty acids, phosphoric acid, glucose
- D. Sphingol, fatty acids, phosphoric acid

Correct Answer: B. Glycerol, fatty acids, phosphoric acid

Biochemical Rationale:

Phosphatidic acid (diacylglycerol 3-phosphate) is the simplest glycerophospholipid and the precursor for all other phosphoglycerides.

Its chemical structure consists of two fatty acids esterified to carbons 1 and 2 of a glycerol backbone, with a phosphate group attached to carbon 3. Complete hydrolytic cleavage yields **glycerol, two fatty acids, and phosphoric acid**.

[24/09/2026 9:27 AM] Imaanaa Harmee:

Is all DNA sequences required for synthesis of particular RNA transcript?

- A. Chromosome

- B. Gene
- C. Alleles
- D. Locus

Correct Answer: B. Gene

Biochemical Rationale:

A **gene** is defined molecularly as the entire linear sequence of genomic DNA necessary to direct the synthesis of a functional RNA transcript or protein. This includes coding regions (exons), noncoding intervening sequences (introns), and associated regulatory elements (promoters, operators, and enhancers).

Which, if any of the following is incorrect about the genetic diseases?

- A. Genetic disease is any of the diseases and disorders that are caused by mutations in one or more genes.
- B. There are about 6000 known genetic disorders; all are fatal or cause severe problems.
- C. Monogenic disorders are inherited disorder due to defect on single gene that produce large effect.
- D. Mitochondrial genetic inheritance disorder is one example of non-classic mendelian inheritance patterns in monogenic disorders.
- E. All are correct

Correct Answer: B. There are about 6000 known genetic disorders; all are fatal or cause severe problems.

Biochemical Rationale:

Statement B is false because while thousands of monogenic and polygenic disorders exist, many are relatively mild, manageable, or asymptomatic under normal conditions (e.g., mild red-green color blindness, benign pentosuria, or alkaptonuria) and are not fatal.

Statements A, C, and D are factually correct: Monogenic disorders stem from mutations in a single gene, and mitochondrial inheritance is a classic non-Mendelian maternal pattern.

[24/09/2026 9:27 AM] Imaanaa Harmee: A large, three generation family in whom multiple members are affected with a rare, undiagnosed disease is being studied. Affected males never produce affected children, but affected females do produce affected children of both sexes when they mate with unaffected males. What is the most likely mode of inheritance?

- A. Autosomal dominant, with expression limited to females
- B. Y-linked
- C. Mitochondrial genetic inheritance disorders
- D. X-linked dominant
- E. X-linked recessive

Correct Answer: C. Mitochondrial genetic inheritance disorders

Genetics Rationale:

Maternal Inheritance: Mitochondria are inherited exclusively from the maternal oocyte because sperm mitochondria are destroyed upon fertilization. Consequently, an affected female passes the mutated mitochondrial genome to 100% of her offspring (both sons and daughters), whereas an affected male never transmits the disease to his children.

A man who is affected with hemophilia A (X-linked recessive) mates with a woman who is a heterozygous carrier of this disorder. What proportion of this couple's daughters will be affected, and what proportion of the daughters will be heterozygous carriers?

(Note: The options for this question appear to be missing, but the question text is complete.)

Correct Answer: 50% of daughters will be affected, and 50% will be heterozygous carriers

Genetics Rationale:

Father (XhY): Contributes an Xh chromosome to all daughters.

Mother (XHXh): Passes the normal XH allele to 50% of her daughters and the mutant Xh allele to the other 50%.

Outcome for Daughters:

XHXh (50%): Heterozygous carriers (phenotypically normal).

XhXh (50%): Homozygous/affected with Hemophilia A.

Heterozygotes for the disease-causing mutation. What is the probability that they will produce phenotypically normal offspring?

A. 50%

B. 25%

C. 75%

D. 100%

E. None

Correct Answer: C. 75%

Genetics Rationale:

For an autosomal recessive trait, a cross between two carrier heterozygotes (Aa×Aa) yields a Mendelian phenotypic ratio of 3 normal : 1 affected:

25% AA: Genotypically and phenotypically normal.

50% Aa: Carrier heterozygotes (phenotypically normal).

25% aa: Affected.

Combining AA (25%) and Aa (50%) gives a 75% probability of producing a phenotypically normal child.

The severe form of alpha-1 antitrypsin deficiency is the result of a single nucleotide substitution that produces a single amino acid substitution. This is best described as a:

A. Frame-shift mutation

B. Large segment deletion mutation

C. Missense mutation

D. Nonsense mutation

E. Silent mutation

Correct Answer: C. Missense mutation

Biochemical Rationale:

A missense mutation is a point mutation in which a single nucleotide base substitution alters a codon to encode a different amino acid (e.g., the Z-allele of α1-antitrypsin results from a point mutation changing Glu342 to Lys).

Other types: A nonsense mutation creates a premature stop codon; a silent mutation does not

change the amino acid; a frame-shift mutation inserts or deletes nucleotides not divisible by three.

Among the following chromosomal abnormalities which one is unusual type of translocation that seen in acrocentric chromosomes where short arm of two chromosomes is lost and long arms are fuse together.

- A. Large segment deletion mutation
- B. Simple translocation
- C. Reciprocal translocation
- D. Robertsonian translocation
- E. Frame shift mutation

Correct Answer: D. Robertsonian translocation

Genetics Rationale:

Robertsonian translocation occurs exclusively between acrocentric chromosomes (chromosomes 13, 14, 15, 21, and 22). The short arms (p-arms) containing redundant ribosomal RNA genes are lost, and the long arms (q-arms) fuse at the centromere to form a single large chromosome. This is a common structural cause of familial Down syndrome (translocation 14/21).

[24/09/2026 9:27 AM] Imaanaa Harmee: Of the following genetic diseases, _____ is autosomal dominant disorder in which genes for low density lipoprotein (LDL) receptor protein at chromosome 19 of one homologous pair is mutant.

- A. Familial hypercholesterolemia
- B. Marfan Syndrome
- C. Hypophosphatemic rickets
- D. Follicular Lymphoma
- E. Burkett's Lymphoma

Correct Answer: A. Familial hypercholesterolemia

Biochemical Rationale:

Familial hypercholesterolemia (FH) is an autosomal dominant metabolic disease caused by loss-of-function mutations in the LDLR gene (located on chromosome 19p13.2). Defective or missing LDL receptors impair clearance of LDL particles from plasma, causing severe hypercholesterolemia, tendon xanthomas, and premature coronary artery disease.

Which of one of the following is true regarding to X-linked disorders:

- A. If heterozygous female met normal male; 100% of female off-springs suffers from disorders in X-linked dominant disorders
- B. If hemizygous male met normal female; 100% of male off-springs suffers from disorders
- C. If hemizygous male met heterozygous female; 50% of male off-springs have normal genotype and phenotype
- D. If heterozygous female met normal male; 100% of female off-springs suffer from disorder in X-linked recessive disorders
- E. None

Correct Answer: C. If hemizygous male met heterozygous female; 50% of male off-springs have normal genotype and phenotype

Genetics Rationale:

Father (XhY): Passes his Y chromosome to all sons.

Mother (XHXh): Passes either her normal XH allele (50%) or her mutant Xh allele (50%) to her sons.

Thus, 50% of the male offspring inherit the normal XH chromosome from their mother and are completely normal in both genotype and phenotype (XHY).

Sex chromosomal Monosomy:

- A. Edward syndrome
- B. Patau syndrome
- C. Down syndrome
- D. Turner Syndrome
- E. Cri-du-chat syndrome

Correct Answer: D. Turner Syndrome

Genetics Rationale:

Turner Syndrome (45,X) is the only viable sex chromosomal monosomy in humans, characterized by the complete or partial absence of a second sex chromosome in females. Down (trisomy 21), Edward (trisomy 18), Patau (trisomy 13), and Cri-du-chat (5p deletion) are all autosomal abnormalities.

Which of the following vitamins or enzymes is unable to protect against free radical damage?

- A. β -Carotene
- B. Glutathione peroxidase
- C. Vitamin E
- D. Vitamin B6
- E. Vitamin C

Correct Answer: D. Vitamin B6

Biochemical Rationale:

Vitamin B6 (pyridoxal phosphate) serves as a coenzyme in amino acid transamination, decarboxylation, and glycogen phosphorylase reactions, but it does not act as a direct antioxidant. Vitamin E (α -tocopherol), Vitamin C (ascorbate), β -carotene, and glutathione peroxidase are primary cellular antioxidants that scavenge reactive oxygen species (ROS).

Superoxide dismutase catalyzes which of the following reactions?

- A. $O_2^{\cdot-} + e^- + 2H^+ \rightarrow H_2O_2$
- B. $2O_2^{\cdot-} + 2H^+ \rightarrow H_2O_2 + O_2$
- C. $O_2^{\cdot-} + HO^- + H^+ \rightarrow CO_2 + H_2O$
- D. $H_2O_2 + O_2 \rightarrow 4H_2O$
- E. $O_2^{\cdot-} + H_2O_2 + H^+ \rightarrow 2H_2O + O_2$

Correct Answer: B. $2O_2^{\cdot-} + 2H^+ \rightarrow H_2O_2 + O_2$

Biochemical Rationale:

Superoxide dismutase (SOD) is a key enzymatic defense against oxidative stress. It catalyzes the dismutation of two superoxide free radicals ($O_2^{\cdot-}$) into hydrogen peroxide (H_2O_2) and molecular oxygen (O_2). The resulting H_2O_2 is subsequently converted to water by catalase or glutathione peroxidase.

[24/09/2026 9:27 AM] Imaanaa Harmee: The mechanism of vitamin E as an antioxidant is best described by which of the following?

- A. Vitamin E binds to free radicals and sequesters them from the contents of the cell.
- B. Vitamin E participates in the oxidation of the radicals.
- C. Vitamin E participates in the reduction of the radicals.
- D. Vitamin E forms a covalent bond with the radicals, thereby stabilizing the radical state.
- E. Vitamin E inhibits enzymes that produce free radicals.

Correct Answer: C. Vitamin E participates in the reduction of the radicals.

Biochemical Rationale:

Vitamin E (α -tocopherol) functions as a major lipid-soluble, chain-breaking antioxidant in cellular membranes. It donates a hydrogen atom (and an electron) to lipid peroxyl radicals ($\text{ROO}\cdot$), thereby reducing the radical into a stable lipid hydroperoxide (ROOH) and halting the chain reaction of lipid peroxidation. The resulting tocopheroxyl radical is relatively stable and is subsequently reduced back to active Vitamin E by Vitamin C (ascorbate).

The level of oxidative damage to mitochondrial DNA is 10 times greater than that to nuclear DNA. This could be due, in part, to which of the following?

- A. Superoxide dismutase is present in the mitochondria.
- B. The nucleus lacks glutathione.
- C. The nuclear membrane presents a barrier to reactive oxygen species.
- D. The mitochondrial membrane is permeable to reactive oxygen species.
- E. Mitochondrial DNA lacks histones.

Correct Answer: E. Mitochondrial DNA lacks histones.

Biochemical Rationale:

Nuclear DNA is tightly wrapped around basic histone octamers to form nucleosomes, which physically shield the DNA backbone from reactive oxygen species (ROS). In contrast, mitochondrial DNA (mtDNA) is circular and lacks protective histones, leaving it directly exposed to high levels of ROS generated as side-products of the mitochondrial electron transport chain (Complexes I and III).

When p53 increases in response to DNA damage, which of the following events occurs?

- A. p53 induces transcription of cdk4.
- B. p53 induces transcription of cyclin D.
- C. p53 binds E2F to activate transcription.
- D. p53 induces transcription of p21.
- E. p53 directly phosphorylates the transcription factor jun.

Correct Answer: D. p53 induces transcription of p21.

Biochemical Rationale:

p53 is a nuclear transcription factor known as the "guardian of the genome." Upon sensing DNA damage, p53 binds to DNA and directly stimulates the transcription of p21 (a Cyclin-Dependent Kinase Inhibitor in the Cip/Kip family). The p21 protein binds and inhibits Cyclin-CDK complexes (such as Cyclin E-CDK2), preventing the phosphorylation of the retinoblastoma protein (Rb) and halting the cell cycle at the G1/S checkpoint to allow DNA repair.

Mutations in the p53 gene are the most common aberrations in spontaneous human cancers. The normal p53 protein affects the cell cycle by:

- A. Inducing cell cycle arrest and apoptosis in response to DNA damage.

- B. Inducing the phosphorylation of Rb in response to mitogens.
- C. Directly inhibiting Cdk inhibitors in response to cell-cell contact and other growth-inhibiting stimuli.
- D. Increasing the activity of cyclin-dependent protein kinases by inducing their phosphorylation.
- E. Binding and thereby inactivating the products of many proapoptotic genes.

Correct Answer: A. Inducing cell cycle arrest and apoptosis in response to DNA damage.

Biochemical Rationale:

Normal p53 prevents damaged or mutated cells from proliferating. Depending on the extent of DNA damage, p53 either halts cell cycle progression at checkpoints (via p21 and GADD45) to allow time for repair, or, if damage is irreparable, upregulates pro-apoptotic genes (such as Bax and IGF-BP3) to induce programmed cell death (apoptosis).

[24/09/2026 9:27 AM] Imaanaa Harmee: Which one the following is not true about M-checkpoint:

- A. Anaphase promoting complex/Cyclosome is key regulator.
- B. Triggered mainly by activation of specific cyclin-Cdk complexes via phosphorylation or dephosphorylation.
- C. Progression of metaphase-to-anaphase transition triggered by cyclical proteolysis.
- D. Checks whether alignment of every chromosome on spindle.
- E. All are true.

Correct Answer: B. Triggered mainly by activation of specific cyclin-Cdk complexes via phosphorylation or dephosphorylation.

Biochemical Rationale:

Why B is INCORRECT: Activation of Cyclin B–CDK1 (MPF) via dephosphorylation by Cdc25 triggers entry into mitosis (the G2/M transition), not the M-checkpoint itself.

M-Checkpoint Mechanism: The M-checkpoint (Spindle Assembly Checkpoint) checks that every chromosome is properly attached to the mitotic spindle at the metaphase plate.

Progression through the metaphase-to-anaphase transition is regulated by the Anaphase-Promoting Complex/Cyclosome (APC/C), which triggers the cyclical proteolysis (degradation) of securin and Cyclin B.

Which one the following incorrect about nucleotides?

- A. They are composed of bases, pentose sugar and phosphate group.
- B. Phosphate group forms ester bond with nucleosides at the 5th or 3rd hydroxyl group of the pentose sugar.
- C. The atoms in bases ring are denoted by prime number.
- D. The atoms in pentose sugar are denoted by prime number.

Correct Answer: C. The atoms in bases ring are denoted by prime number.

Biochemical Rationale:

Nomenclature Convention: To distinguish the carbon atoms of the pentose sugar from those of the nitrogenous ring, the carbons in the pentose sugar are specifically designated with prime numbers (1',2',3',4',5'). The atoms in the heterocyclic base ring (purines or pyrimidines) are designated by unprimed standard integers (1,2,3...).

Which of the following is true about DNA polymerase?

- A. It can synthesize DNA in the 5' to 3' direction.
- B. It can synthesize DNA in the 3' to 5' direction.
- C. It can synthesize mRNA in the 3' to 5' direction.
- D. It can synthesize mRNA in the 5' to 3' direction.

Correct Answer: A. It can synthesize DNA in the 5' to 3' direction.

Biochemical Rationale:

All DNA polymerases require a free 3'-OH group and synthesize new DNA strands exclusively in the 5'→3' direction, reading the complementary template strand in the antiparallel 3'→5' direction. (RNA polymerases, not DNA polymerases, synthesize mRNA).

Which of the following reactions is required for proofreading during DNA replication by DNA polymerase?

Correct Answer / Explanation: 3'→5' Exonuclease activity (Hydrolytic cleavage of mismatched 3'-terminal nucleotides)

Biochemical Rationale:

Replicative DNA polymerases (such as DNA Polymerase III in prokaryotes, or Pol δ / Pol ε in eukaryotes) possess an intrinsic 3'→5' exonuclease activity. When an incorrect base is added to the growing 3'-end, the enzyme pauses, shifts backward, and hydrolytically excises the mismatched 3'-terminal nucleotide before resuming 5'→3' polymerase synthesis.

Which of the following enzymes remove supercoiling in replicating DNA ahead of the replication fork?

- A. DNA polymerases
- B. Helicases
- C. Primases
- D. Topoisomerases

Correct Answer: D. Topoisomerases

Biochemical Rationale:

As DNA helicase separates the parental strands, positive supercoils (torsional strain) accumulate in the double helix ahead of the advancing replication fork. DNA Topoisomerases (Type I and Type II / DNA Gyrase) break phosphodiester bonds in one or both strands, allow the helix to unwind to relieve supercoiling, and reseal the DNA backbone.

DNA unwinding is done by _____.

- A. Ligase
- B. Helicase
- C. Topoisomerase
- D. Exonuclease

Correct Answer: B. Helicase

Biochemical Rationale:

DNA Helicase (such as DnaB in E. coli) binds to single-stranded DNA near the replication fork and uses energy derived from ATP hydrolysis to break hydrogen bonds between complementary base pairs, unwinding the parental double helix.

[24/09/2026 9:27 AM] Imaanaa Harmee: **Transcription is catalyzed by _____.**

- A. DNA-dependent RNA polymerases

- B. Reverse transcriptase
- C. RNA-dependent DNA polymerases
- D. DNA ligases

Correct Answer: A. DNA-dependent RNA polymerases

Rationale: Transcription is the process of synthesizing an RNA molecule using a DNA template. The enzyme responsible is DNA-dependent RNA polymerase, which reads the antisense DNA template strand in the 3'→5' direction to synthesize a complementary RNA transcript in the 5'→3' direction.

RNA primer is removed from the Okazaki fragment by ____.

- A. DNA polymerase I
- B. DNA polymerase II
- C. DNA polymerase III
- D. RNA polymerase

Correct Answer: A. DNA polymerase I

Rationale: In prokaryotes (E. coli), DNA Polymerase I possesses a unique 5'→3' exonuclease activity that hydrolytically removes the RNA primer ahead of it while simultaneously synthesizing DNA (5'→3' polymerase activity) to fill the gap between Okazaki fragments.

Which of the following solves the problem of shortening of chromosomes in each cell cycle?

- A. DNA ligase
- B. Endonuclease
- C. Telomerase
- D. RNA polymerase

Correct Answer: C. Telomerase

Rationale: Linear eukaryotic chromosomes experience the "end-replication problem," where the removal of terminal RNA primers on lagging strands leaves a gap that causes chromosomes to shorten with every round of replication. Telomerase is a specialized ribonucleoprotein (reverse transcriptase) carrying its own RNA template that synthesizes non-coding hexameric repeats (TTAGGG) at the 3'-ends of chromosomes, preventing loss of vital genetic code.

Which of the following enzyme is not required for the synthesis of lagging strand?

- A. Primase
- B. Gyrase
- C. Helicase
- D. Exonuclease

Correct Answer: D. Exonuclease (or standalone exonucleases)

Rationale: Lagging strand synthesis is discontinuous and requires Helicase (unwinding), Gyrase/Topoisomerase (relieving torsional strain), and Primase (synthesizing RNA primers for each Okazaki fragment). While primer removal requires the 5'→3' exonuclease activity intrinsic to DNA Polymerase I, standalone exonuclease enzymes are not separate structural requirements for assembling lagging strand fragments.

Which of the following is an incorrect statement about mRNA?

- A. Cap is added to the 5' end
- B. Exons are removed and introns are joined together
- C. Introns are removed and exons are joined together
- D. Poly-A tail is added to the 3' end

Correct Answer: B. Exons are removed and introns are joined together

Rationale: Why B is FALSE: During post-transcriptional pre-mRNA splicing, non-coding intervening sequences (introns) are removed, and coding sequences (exons) are joined (spliced) together to form mature mRNA. Statements A, C, and D are true: Eukaryotic mRNA receives a 7-methylguanosine cap at its 5'-end, has its introns spliced out, and receives a poly-A tail at its 3'-end.

Which one of the serves as a starter sequence for DNA polymerase III?

- A. Promoter
- B. RNA primer
- C. Starting codon
- D. All of the above

Correct Answer: B. RNA primer

Rationale: All DNA polymerases require a pre-existing free 3'-hydroxyl (–OH) group to initiate nucleotide addition. Primase (an RNA polymerase) synthesizes a short complementary RNA primer that provides this mandatory free 3'-OH starter sequence for DNA Polymerase III.

Unwound DNA prevented from degradation and re-annealing by _____.

- A. Double strand binding protein
- B. Helicase
- C. Single strand binding protein
- D. All of the above

Correct Answer: C. Single strand binding protein

Rationale: As DNA helicase unwinds the double helix, Single-Stranded Binding Proteins (SSBs) coat the exposed single-stranded template DNA. SSBs prevent the two complementary strands from re-annealing (spontaneously reforming double-stranded DNA) and shield single strands from nucleolytic degradation.

[24/09/2026 9:27 AM] Imaanaa Harmee: **The 20 different amino acids are in the protein are normally coded by:**

- A. 60 codons
- B. 61 codons
- C. 63 codons
- D. 64 codons

Correct Answer: B. 61 codons

Rationale: The standard genetic code consists of $4^3 = 64$ triplet codons. Three of these (UAA, UAG, UGA) are stop (nonsense) codons that terminate translation and do not code for any

amino acid. The remaining 61 sense codons specify the 20 standard amino acids incorporated into proteins.

Codon and anti-codon are located in:

- A. tRNA and mRNA respectively
- B. mRNA and tRNA respectively
- C. tRNA and mRNA respectively
- D. tRNA and mRNA respectively

(Note: Options A, C, and D are identical and seem to be a copying error.)

Correct Answer: B. mRNA and tRNA respectively

Rationale: The codon is a 3-nucleotide coding sequence located on the messenger RNA (mRNA) transcript. The anticodon is a complementary 3-nucleotide triplet located on the anticodon loop of the transfer RNA (tRNA) adapter molecule.

In the translation process, ribosomes on mRNA moves from one codon to another one codon and the peptidyl-tRNA is moved from the A-site to the P-site by:

- A. Translocase
- B. Peptidyl transferase
- C. 50S ribose subunit
- D. 30S ribose subunit

Correct Answer: A. Translocase

Rationale: During the elongation phase of translation, Translocase (Elongation Factor EF-G in prokaryotes / EF-2 in eukaryotes) uses energy from GTP hydrolysis to advance the ribosome along mRNA by one codon (5'→3' direction), moving the uncharged tRNA to the E-site and shifting the peptidyl-tRNA from the A-site to the P-site.

[24/09/2026 9:27 AM] Imaanaa Harmee: In the translation process, the first incoming of fMet-tRNA molecules bind to _____

- A. A-site
- B. P-site
- C. E-site
- D. All of the above

Correct Answer: B. P-site

Biochemical Rationale:

During translational initiation in prokaryotes, the special initiating fMet-tRNA molecule enters directly into the P (peptidyl) site of the ribosome. All subsequent incoming aminoacyl-tRNAs enter at the A (aminoacyl) site during the elongation phase.

Which of the following lipid is used for energy storage?

- A. Glycolipid
- B. Glycerophospholipid
- C. Sphingolipids
- D. Triacylglycerol

Correct Answer: D. Triacylglycerol

Biochemical Rationale:

Triacylglycerols (triglycerides) consist of three fatty acids esterified to a glycerol backbone.

Because they are completely uncharged and hydrophobic, they pack densely without bound water in adipose tissue lipid droplets, serving as the body's primary long-term metabolic fuel reservoir.

Phospholipids are important cell membrane components because:

- A. They have both polar and nonpolar portions
- B. They have glycerol
- C. They can form bilayers in water
- D. They combine covalently with proteins

Correct Answer: A. They have both polar and nonpolar portions (or C. They can form bilayers in water)

Biochemical Rationale:

Phospholipids are amphipathic molecules containing a polar hydrophilic head group and nonpolar hydrophobic fatty acyl tails. This dual property causes them to spontaneously assemble into continuous lipid bilayers in aqueous cellular environments.

The repeating unit in hyaluronic acid is:

- A. Glycerol and galactose
- B. Glycerol and N-acetyl galactosamine
- C. Glycerol and glucosamine
- D. Glycerol and N-acetyl glucosamine

Correct Answer: D. Glycerol and N-acetyl glucosamine

(Biochemical composition: D-Glucuronic acid + N-acetylglucosamine)

Biochemical Rationale:

Hyaluronic acid is an unbranched glycosaminoglycan (GAG) composed of repeating disaccharides of D-glucuronic acid and N-acetylglucosamine (GlcNAc). (Note: "Glycerol" in choice D represents a common typographical error for "Glucuronic acid" in standard test banks).

Glucose and galactose are epimers and they differ structurally in orientation of H and OH on:

- A. C4
- B. C3
- C. C2
- D. C5

Correct Answer: A. C4

Biochemical Rationale:

Epimers are stereoisomers that differ in configuration around only one specific asymmetric carbon. D-glucose and D-galactose differ strictly in the orientation of the hydroxyl (–OH) group at carbon 4 (C-4).

The reducing ability of a carbohydrate is due to the presence/or formation of:

- A. A free carboxyl group
- B. A free hydroxyl groups

- C. Presence of a 1-4 linkage
- D. Formation of ring structure

Correct Answer: B. A free hydroxyl groups (specifically the free anomeric –OH group)

Biochemical Rationale:

A carbohydrate exhibits reducing properties if it possesses an unlinked, free anomeric carbon with a free hydroxyl group. In solution, this permits mutarotation and ring opening to expose a free aldehyde or ketone group that donates electrons to reduce mild oxidizing agents.

Phosphatidic acid on hydrolysis yields:

- A. Glycerol, fatty acids, phosphoric acid, choline
- B. Glycerol, fatty acids, phosphoric acid
- C. Glycerol, fatty acids, phosphoric acid, glucose
- D. Sphingol, fatty acids, phosphoric acid

Correct Answer: B. Glycerol, fatty acids, phosphoric acid

Biochemical Rationale:

Phosphatidic acid (diacylglycerol 3-phosphate) consists of two fatty acyl chains esterified to carbons 1 and 2 of glycerol, with a phosphate group at carbon 3. Complete hydrolytic cleavage yields glycerol, two fatty acids, and phosphoric acid.

Is all DNA sequences required for synthesis of particular RNA transcript?

- A. Chromosome
- B. Gene
- C. Alleles
- D. Locus

Correct Answer: B. Gene

Biochemical Rationale:

A gene is defined as the complete region of genomic DNA required to direct the synthesis of a functional RNA transcript or protein product, including coding regions (exons), non-coding introns, and associated regulatory sequences.

[24/09/2026 9:27 AM] Imaanaa Harmee: Which, if any of the following is incorrect about the genetic diseases?

- A. Genetic disease is any of the diseases and disorders that are caused by mutations in one or more genes.
- B. There are about 6000 known genetic disorders; all are fatal or cause severe problems.
- C. Monogenic disorders are inherited disorder due to defect on single gene that produce large effect.
- D. Mitochondrial genetic inheritance disorder is one example of non-classic mendelian inheritance patterns in monogenic disorders.
- E. All are correct

Correct Answer: B. There are about 6000 known genetic disorders; all are fatal or cause severe problems.

Biochemical Rationale:

Statement B is false because many genetic variations and single-gene traits are benign, mild, or clinically manageable (e.g., red-green color blindness or benign pentosuria) rather than universally fatal.

Exam: Biochemistry

Year: 2016

Dr. Doc Tutorial

[24/09/2026 9:27 AM] Imaanaa Harmee: Which one of the following is the C-2 epimer of D-glucose?

- A. D-mannose
- B. D-galactose
- C. D-fructose
- D. D-ribose

Correct Answer: A. D-mannose

Biochemical Rationale:

Epimers are stereoisomers that differ in configuration around only a single asymmetric carbon atom. D-mannose differs from D-glucose strictly at carbon 2 (C-2), making it the C-2 epimer. (In contrast, D-galactose is the C-4 epimer of D-glucose).

Which one of the following is the sugar alcohol of galactose?

- A. Glucitol
- B. Mannitol
- C. Dulcitol
- D. Xylitol

Correct Answer: C. Dulcitol

Biochemical Rationale:

Reduction of the aldehyde group of D-galactose yields the polyol dulcitol (also known as galactitol), which accumulates in the lens of the eye in patients with galactosemia. (Glucitol/sorbitol derives from glucose, mannitol from mannose, and xylitol from xylose).

The correct formula of arachidonic acid is:

- A. 5,8,11,14-eicosatetraenoic acid, ω -6
- B. 9,12-octadecadienoic acid, ω -6
- C. 9,12,15-octadecatrienoic acid, ω -3
- D. 20:5 ω -3 (EPA)

Correct Answer: A. 5,8,11,14-eicosatetraenoic acid, ω -6

Biochemical Rationale:

Arachidonic acid is a 20-carbon polyunsaturated fatty acid (PUFA) containing four cis double bonds located at positions 5, 8, 11, and 14. Its systematic shorthand is 20:4 (5,8,11,14),

belonging to the ω -6 (n-6) family because its final double bond is 6 carbons from the terminal methyl (ω) end.

Which one of the following fatty acids is precursor of eicosanoids?

- A. Arachidonic acid
- B. Palmitic acid
- C. Stearic acid
- D. Oleic acid

Correct Answer: A. Arachidonic acid

Biochemical Rationale:

Arachidonic acid released from membrane phospholipids by Phospholipase A₂ serves as the primary substrate for eicosanoid synthesis via the cyclooxygenase (COX) pathway (yielding prostaglandins and thromboxanes) and the lipoxygenase (LOX) pathway (yielding leukotrienes).

Niemann–Pick disease is caused by deficiency of:

- A. Sphingomyelinase
- B. Hexosaminidase A
- C. Glucocerebrosidase
- D. Galactocerebrosidase

Correct Answer: A. Sphingomyelinase

Biochemical Rationale:

Niemann–Pick disease (Types A and B) is an autosomal recessive lysosomal storage disorder caused by a deficiency of acid sphingomyelinase, leading to pathological accumulation of sphingomyelin in reticuloendothelial lysosomal compartments (hepato-splenomegaly and neurodegeneration).

Lung surfactant complexes contain dipalmitoyl-_____:

- A. Sphingomyelin
- B. Lecithin (phosphatidylcholine)
- C. Cardiolipin
- D. Ceramide

Correct Answer: B. Lecithin (phosphatidylcholine)

Biochemical Rationale:

Dipalmitoylphosphatidylcholine (DPPC), commonly called dipalmitoyl lecithin, is the major lipid component (~80%) of pulmonary surfactant secreted by type II pneumocytes, lowering surface tension at the alveolar air-fluid interface.

Which one of the following is a component of cellular and subcellular membranes?

- A. Cholesterol
- B. Sphingomyelin
- C. Phosphatidylserine
- D. Cardiolipin

Correct Answer: D. Cardiolipin (or A/B/C/D depending on emphasis on subcellular organelles)

Biochemical Rationale:

While cholesterol, sphingomyelin, and phosphatidylserine are predominantly concentrated in the plasma membrane, cardiolipin (diphosphatidylglycerol) is uniquely localized within the inner mitochondrial membrane (a key subcellular organelle membrane), where it stabilizes respiratory chain complexes I–IV.

The first criminal caught based on DNA fingerprinting evidence was:

- A. Colin Pitchfork
- B. Jeffrey Dahmer
- C. Ted Bundy
- D. Charles Whitman

Correct Answer: A. Colin Pitchfork

Biochemical Rationale:

In 1987, Colin Pitchfork became the first person in history to be convicted of murder using DNA profiling/fingerprinting evidence developed by Sir Alec Jeffreys at the University of Leicester.

[24/09/2026 9:27 AM] Imaanaa Harmee: RNA used for gene knockout in cancer cells:

- A. siRNA
- B. rRNA
- C. tRNA
- D. snRNA

Correct Answer: A. siRNA

Biochemical Rationale:

Small interfering RNA (siRNA) is a short (~21–23 bp) double-stranded non-coding RNA that incorporates into the RISC complex to direct sequence-specific mRNA degradation, making it a key tool for targeted gene silencing and oncogene knockout.

The end of DNA is called:

- A. Telomere
- B. Centromere
- C. Histone tail
- D. Satellite DNA

Correct Answer: A. Telomere

Biochemical Rationale:

Telomeres are specialized, non-coding hexameric repeat sequences (TTAGGG_n) located at the terminal physical ends of linear eukaryotic chromosomes, protecting them from degradation and end-to-end fusion.

[24/09/2026 9:27 AM] Imaanaa Harmee: Turner's syndrome karyotype is:

- A. 45,XO
- B. 47,XXY
- C. 47,XXX
- D. 46,XY

Correct Answer: A. 45,XO

Genetics Rationale:

Turner syndrome is a sex chromosome monosomy in females characterized by the partial or complete loss of one X chromosome, yielding a 45,X (or 45,XO) karyotype.

Klinefelter's syndrome karyotype is:

- A. 45,XO
- B. 47,XXY
- C. 47,XYY
- D. 46,XY

Correct Answer: B. 47,XXY

Genetics Rationale:

Klinefelter syndrome is a male sex chromosome aneuploidy caused by nondisjunction that results in an extra X chromosome (47,XXY).

Jacob's syndrome is:

- A. 45,XO
- B. 47,XXY
- C. 47,XYY
- D. 46,XY

Correct Answer: C. 47,XYY

Genetics Rationale:

Jacob's syndrome (also known as XYY syndrome) is a male sex chromosome variation caused by paternal non-disjunction during meiosis II, yielding an extra Y chromosome (47,XYY).

Microscope-based tests include:

- A. Cytogenetics
- B. Karyotyping
- C. FISH
- D. All of the above

Correct Answer: D. All of the above

Genetics Rationale:

Classical cytogenetics, G-banded karyotyping, and Fluorescence In Situ Hybridization (FISH) all require specialized light or fluorescence microscopy to inspect metaphase chromosomes or fluorescently labeled DNA probes.

Vacuoles which pump excess water out of the cell:

- A. Food vacuoles
- B. Contractile vacuoles
- C. Central vacuoles
- D. Lysosomes

Correct Answer: B. Contractile vacuoles

Cell Biology Rationale:

Contractile vacuoles are specialized organelles present in freshwater protists and unicellular

organisms that collect and rhythmically pump excess water out of the cell to prevent lysis and maintain osmoregulation.

In which organelle is tRNA transcribed/processed?

- A. Nucleolus
- B. Nucleus
- C. Ribosome
- D. Cytoplasm

Correct Answer: B. Nucleus

Biochemical Rationale:

Transfer RNA (tRNA) genes are transcribed in the nucleus by RNA Polymerase III. The primary tRNA transcripts undergo 5' leader trimming, 3' CCA addition, base modification, and intron splicing within the nucleus prior to nuclear export. (Note: The nucleolus is the site of ribosomal RNA transcription and ribosome subunit assembly).

Absence of chromosome 18 results in:

- A. Edwards syndrome
- B. Jacob's syndrome
- C. Cri-du-chat
- D. Patau syndrome

Correct Answer: A. Edwards syndrome

Genetics Rationale:

Edwards syndrome is caused by an autosomal chromosome abnormality involving chromosome 18—specifically Trisomy 18 (47,XX/XY,+18). (Patau syndrome is Trisomy 13; Down syndrome is Trisomy 21).

Note: The question says "absence" but the correct association is with an extra copy (trisomy) of chromosome 18. The intended answer remains Edwards syndrome.

Absence of chromosome 21 results in:

- A. Edwards syndrome
- B. Jacob's syndrome
- C. Down syndrome
- D. Cri-du-chat

Correct Answer: C. Down syndrome

Genetics Rationale:

Down syndrome is the most frequent human autosomal chromosome disorder, caused by Trisomy 21 (47,XX/XY,+21) or a Robertsonian translocation involving chromosome 21.

Note: The question says "absence" but the correct association is with an extra copy (trisomy) of chromosome 21. The intended answer remains Down syndrome.

Chromosome 5p deletion results in:

- A. Edwards syndrome
- B. Jacob's syndrome
- C. Down syndrome
- D. Cri-du-chat

Correct Answer: D. Cri-du-chat

Genetics Rationale:

Cri-du-chat ("cry of the cat") syndrome is a partial deletion syndrome caused by a structural loss of the short arm of chromosome 5 (5p deletion / 5p-).

Which one of the following is X-linked recessive?

- A. Color blindness
- B. Hemophilia A
- C. Duchenne muscular dystrophy
- D. All of the above

Correct Answer: D. All of the above

Genetics Rationale:

Red-green color blindness, Hemophilia A (Factor VIII deficiency), and Duchenne muscular dystrophy (dystrophin gene mutation) are classic textbook examples of X-linked recessive disorders.

[24/09/2026 9:27 AM] Imaanaa Harmee: Which one of the following is X-linked dominant?

- A. Alport syndrome
- B. Vitamin D-resistant rickets
- C. Rett syndrome
- D. All of the above

Correct Answer: D. All of the above

Genetics Rationale:

Alport syndrome (classic form caused by COL4A5 mutations), Vitamin D-resistant rickets (Hypophosphatemic rickets caused by PHEX mutations), and Rett syndrome (caused by MECP2 mutations) are all classic textbook examples of X-linked dominant inheritance.

Which one of the following is caused by trisomy 13?

- A. Patau syndrome
- B. Down's syndrome
- C. Edwards syndrome
- D. Cri-du-chat

Correct Answer: A. Patau syndrome

Genetics Rationale:

Patau syndrome is caused by an extra copy of chromosome 13 (Trisomy 13: 47,XX/XY,+13). Comparison: Down syndrome is Trisomy 21, Edwards syndrome is Trisomy 18, and Cri-du-chat is caused by a 5p deletion.

Which one of the following enzymes is used in xenobiotic metabolism?

- A. Glutathione (GSH)
- B. Peroxidase
- C. Catalase
- D. Superoxide dismutase

Correct Answer: A. Glutathione (GSH) (specifically Glutathione S-transferases)

Biochemical Rationale:

Glutathione (GSH) plays a central role in Phase II xenobiotic biotransformation. Enzymes known as glutathione S-transferases (GSTs) conjugate reduced glutathione to electrophilic xenobiotics, making toxic compounds water-soluble for renal or biliary excretion.

Who proposed the induced fit model of enzyme action?

- A. Koshland
- B. Fischer
- C. Michaelis
- D. Lineweaver

Correct Answer: A. Koshland

Biochemical Rationale:

Daniel Koshland proposed the Induced Fit model in 1958, which states that the active site of an enzyme is flexible and undergoes dynamic conformational shifts upon substrate binding to optimize catalytic group orientation. (Emil Fischer proposed the rigid "Lock and Key" model).

Which one of the following is an example of competitive inhibitor?

- A. Statins (HMG-CoA reductase)
- B. Cyanide (cytochrome oxidase)
- C. Aspirin (COX enzyme)
- D. Allopurinol (xanthine oxidase)

Correct Answer: A. Statins (HMG-CoA reductase)

Biochemical Rationale:

Statins (e.g., atorvastatin, simvastatin) are structural analogues of 3-hydroxy-3-methylglutaryl-CoA (HMG-CoA). They act as reversible competitive inhibitors of HMG-CoA reductase, competing directly with HMG-CoA for binding at the catalytic active site to reduce cholesterol synthesis.

Comparison: Aspirin is an irreversible inhibitor of COX, and Cyanide is a non-competitive/cytochrome oxidase inhibitor.

AST (Aspartate transaminase) is mainly used for diagnosis of:

- A. Liver disease
- B. Kidney disease
- C. Pancreatitis
- D. Muscular dystrophy

Correct Answer: A. Liver disease

Biochemical Rationale:

AST (Aspartate aminotransferase) is a key transaminase enzyme present in high concentrations in hepatocytes (both cytosol and mitochondria). Elevated serum AST levels serve as a cardinal diagnostic biomarker for hepatocellular damage and liver disease (as well as myocardial infarction and muscle trauma).

ALT (Alanine transaminase) is mainly used for diagnosis of:

- A. Liver disease
- B. Kidney disease
- C. Myocardial infarction
- D. Hemolysis

Correct Answer: A. Liver disease

Biochemical Rationale:

ALT (Alanine aminotransferase) is localized primarily in the cytoplasm of liver cells. Because it is much more liver-specific than AST, an elevation in serum ALT is the primary laboratory hallmark of acute and chronic liver disease (e.g., viral hepatitis, toxic liver injury).

LCT proteins are associated with:

- A. Kinesin and Dynein
- B. Actin filaments
- C. Collagen fibers
- D. Tubulin dimers

Correct Answer: A. Kinesin and Dynein (or D. Tubulin dimers)

Biochemical Rationale:

Light Chain Tectonic/Tower (LCT) / Light Chains are integral subunit components of motor protein complexes (kinesin and dynein) that facilitate cargo transport along microtubule tracks constructed from tubulin dimers.

[24/09/2026 9:27 AM] Imaanaa Harmee: Which one of the following proteins is used for cargo transport?

- A. Kinesin
- B. Dynein
- C. Myosin
- D. All of the above

Correct Answer: D. All of the above

Biochemical Rationale:

Kinesin transports intracellular cargo toward the plus-end of microtubules (anterograde). Dynein transports cargo toward the minus-end of microtubules (retrograde). Myosin (such as Myosin V) transports cellular cargo along actin microfilaments.

Which of the following proteins maintains plasma oncotic (colloid osmotic) pressure?

- A. Albumin
- B. Globulin
- C. Fibrinogen
- D. Transferrin

Correct Answer: A. Albumin

Biochemical Rationale:

Serum albumin is the most abundant plasma protein (~3.5–5.0 g/dL) synthesized by the liver. Due to its high concentration and molecular weight, albumin generates approximately 75% to 80% of the total plasma colloid osmotic (oncotic) pressure, preventing fluid extravasation into interstitial spaces (edema).

[24/09/2026 9:27 AM] Imaanaa Harmee: Deficiency of α_1 -antitrypsin causes:

- A. Emphysema

- B. Hemophilia
- C. Sickle cell anemia
- D. Asthma

Correct Answer: A. Emphysema

Biochemical Rationale:

α 1-Antitrypsin (α 1-AT, a serine protease inhibitor) normally inhibits neutrophil elastase in alveolar tissue. Inherited α 1-AT deficiency leaves elastase unopposed to destroy the elastin fibers of the alveolar walls, leading to progressive pulmonary emphysema.

Dough extensibility of cereals is due to:

- A. Gluten
- B. Gliadin
- C. Glutenin
- D. Amylopectin

Correct Answer: B. Gliadin

Biochemical Rationale:

Wheat gluten is composed of two primary storage protein fractions: gliadin and glutenin. Gliadin provides monomeric viscous flow and extensibility (stretchability) to cereal dough, whereas glutenin provides elasticity and tensile strength.

Dough elasticity of cereals is due to:

- A. Glutenin
- B. Gliadin
- C. Gluten
- D. Amylose

Correct Answer: A. Glutenin

Biochemical Rationale:

Glutenin forms massive high-molecular-weight protein polymers stabilized by extensive interchain disulfide bonds. This polymeric network gives dough its elasticity and resistance to deformation during baking.

α -Keratin is rich in which amino acid?

- A. Cysteine
- B. Methionine
- C. Tryptophan
- D. Tyrosine

Correct Answer: A. Cysteine

Biochemical Rationale:

α -Keratin is a fibrous structural protein present in skin, hair, and nails that contains an unusually high concentration of cysteine residues. These cysteines form covalent interchain disulfide bonds ($-S-S-$) that impart mechanical toughness and rigidity to the protein matrix.

Which type of collagen is most common in children (fetal skin & blood vessels)?

- A. Type I
- B. Type II
- C. Type III
- D. Type IV

Correct Answer: C. Type III

Biochemical Rationale:

Type III collagen (reticular collagen) forms thin, extensible fibers prevalent in distensible vascular walls, internal visceral organs, and fetal/embryonic skin. During postnatal maturation and wound remodeling, much of it is gradually replaced by high-tensile Type I collagen.

Which amino acid is present in desmosine?

- A. Lysine
- B. Proline
- C. Hydroxyproline
- D. Glycine

Correct Answer: A. Lysine

Biochemical Rationale:

Desmosine is a specialized tetra-functional cross-linking structure unique to elastin. It is formed extracellularly from four lysine residues (specifically three allysine aldehydes generated by lysyl oxidase plus one unaltered lysine residue) to link tropoelastin chains into a rubbery network.

Which amino acid is most abundant in collagen?

- A. Glycine
- B. Proline
- C. Hydroxyproline
- D. Alanine

Correct Answer: A. Glycine

Biochemical Rationale:

Glycine constitutes approximately one-third (33%) of all amino acid residues in collagen. It occurs at every third position of the polypeptide chain (–Gly–X–Y–) because its single hydrogen atom side chain is the only residue small enough to fit into the restricted central space where the three chains meet in the triple helix.

The first protein to cut during reparation of the damaged DNA by base excision repair system is:

- A. Peptidase
- B. Glycosylase
- C. AP endonuclease
- D. DNA ligase

Correct Answer: B. Glycosylase

Biochemical Rationale:

Base Excision Repair (BER) is initiated by a specific DNA glycosylase. The glycosylase recognizes damaged or abnormal bases (e.g., uracil formed by cytosine deamination) and hydrolytically cleaves the N-glycosidic bond joining the base to the sugar-phosphate backbone, leaving an AP (apurinic/apyrimidinic) site.

In base excision repair, after DNA glycosylase removes the damaged base, the next enzyme that acts is:

- A. DNA ligase
- B. AP endonuclease
- C. DNA polymerase α
- D. Topoisomerase I

Correct Answer: B. AP endonuclease

Biochemical Rationale:

After a DNA glycosylase excises the damaged base to create an abasic (AP) site, an AP endonuclease recognizes the missing base and makes a 5' phosphodiester cut adjacent to the AP site. Deoxyribose phosphate lyase then removes the sugar-phosphate remnant, allowing DNA polymerase and DNA ligase to fill and seal the gap.

[24/09/2026 9:27 AM] Imaanaa Harmee: Which one of the following disorders is most strongly associated with defective nucleotide excision repair?

- A. Xeroderma pigmentosum
- B. Ataxia telangiectasia
- C. Bloom syndrome
- D. Fanconi anemia

Correct Answer: A. Xeroderma pigmentosum

Biochemical Rationale:

Xeroderma pigmentosum (XP) is an autosomal recessive genetic disease caused by defects in XP proteins required for Nucleotide Excision Repair (NER). Affected individuals cannot excise UV-induced pyrimidine (thymine) dimers, resulting in severe sun sensitivity and a high incidence of early skin malignancies.

[24/09/2026 9:27 AM] Imaanaa Harmee: BRCA1 and BRCA2 proteins are primarily involved in:

- A. Base excision repair
- B. Mismatch repair
- C. Homologous recombination repair of double-strand breaks
- D. Non-homologous end joining

Correct Answer: C. Homologous recombination repair of double-strand breaks

Biochemical Rationale:

BRCA1 and BRCA2 are key tumor suppressor proteins required for high-fidelity, error-free repair of double-strand DNA breaks via homologous recombination (HR) during the S and G2 phases of the cell cycle. Germline loss-of-function mutations in these genes lead to genomic instability and a significantly increased risk of hereditary breast and ovarian cancers.

Which tumor suppressor gene product directly induces transcription of p21 to halt cell cycle progression?

- A. Rb
- B. BRCA2
- C. p53
- D. APC

Correct Answer: C. p53

Biochemical Rationale:

Upon sensing DNA strand breaks or genomic stress, the transcription factor p53 upregulates expression of the p21 gene. The p21 protein is a cyclin-dependent kinase inhibitor (CKI) that binds and inactivates Cyclin E-CDK2 and Cyclin D-CDK4/6 complexes, maintaining the retinoblastoma protein (Rb) in its hypophosphorylated (active) state to freeze the cell cycle at the G1/S checkpoint.

Which checkpoint ensures that every chromosome is attached to the spindle before anaphase begins?

- A. G1/S checkpoint
- B. G2/M checkpoint
- C. Metaphase (spindle assembly) checkpoint
- D. Cytokinesis checkpoint

Correct Answer: C. Metaphase (spindle assembly) checkpoint

Biochemical Rationale:

The Metaphase (Spindle Assembly) Checkpoint monitors whether all sister chromatids are properly kinetochore-attached to mitotic spindle microtubules and aligned at the metaphase plate. Unattached kinetochores inhibit the Anaphase-Promoting Complex/Cyclosome (APC/C), preventing premature securin degradation and chromosome nondisjunction.

An increase in the activity of which enzyme will increase free radical generation and oxidative stress?

- A. NADPH oxidase
- B. Superoxide dismutase
- C. Glutathione peroxidase
- D. Catalase

Correct Answer: A. NADPH oxidase

Biochemical Rationale:

NADPH oxidase (NOX) is a membrane-bound enzyme complex that transfers electrons from NADPH to molecular oxygen to intentionally produce superoxide radicals ($O_2^{\cdot-}$). It plays a physiological role in the phagocytic respiratory burst to destroy micro-organisms, but pathologically increases cellular oxidative stress when overactivated.

Contrast: Superoxide dismutase, catalase, and glutathione peroxidase are protective antioxidant enzymes that degrade reactive oxygen species.

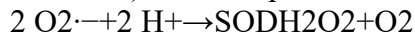
Superoxide dismutase converts superoxide radicals into:

- A. Oxygen and hydrogen peroxide
- B. Hydroxyl radicals
- C. Water only
- D. Singlet oxygen

Correct Answer: A. Oxygen and hydrogen peroxide

Biochemical Rationale:

Superoxide dismutase (SOD) catalyzes the dismutation (simultaneous oxidation and reduction) of two superoxide anions into hydrogen peroxide and molecular oxygen:



The generated H_2O_2 is subsequently detoxified into water by catalase or glutathione peroxidase.

Which vitamin does not act as an antioxidant against free radical damage?

- A. Vitamin E
- B. Vitamin C
- C. Vitamin B6
- D. β -carotene

Correct Answer: C. Vitamin B6

Biochemical Rationale:

Vitamin B6 (pyridoxal phosphate/PLP) functions as an essential coenzyme in nitrogen metabolism—catalyzing transamination, decarboxylation, and glycogen phosphorylase reactions—rather than serving as a direct antioxidant.

Contrast: Vitamin E (membrane chain-breaking lipid antioxidant), Vitamin C (water-soluble radical scavenger), and β -carotene (singlet oxygen quencher) are major biological antioxidants.

[24/09/2026 9:32 AM] Imaanaa Harmee: The mechanism of competitive inhibition is characterized by:

- A. Increase in K_m , no change in V_{max}
- B. Decrease in K_m , no change in V_{max}
- C. Decrease in V_{max} , no change in K_m
- D. Decrease in both K_m and V_{max}

Correct Answer: A. Increase in K_m , no change in V_{max}

Biochemical Rationale:

A competitive inhibitor structurally resembles the substrate and binds reversibly to the free enzyme's active site. Because it competes with the substrate for active site occupancy, higher substrate concentrations are required to reach half-maximal velocity, increasing the apparent K_m . However, at saturating substrate concentrations, the inhibitor is completely displaced, so the maximal velocity ($V_{max}^{[f_0]}$) remains unchanged.

Exam: biochemistry

Year: 2018 E.C

Dr. Doc Tutorials

[24/09/2026 9:32 AM] Imaanaa Harmee: All are true except:

- A. A fatty acid esterified with an alcohol of high molecular weight is wax
- B. Higher alcohol present in wax is cetyl
- C. Gangliosides are complex glycosphingolipids found in brain
- D. None

Correct Answer: D. None (All statements A, B, and C are true)

Biochemical Rationale:

Statement A is true: Waxes are chemically defined as esters formed between long-chain fatty acids and high-molecular-weight monohydric alcohols.

Statement B is true: High-molecular-weight aliphatic monohydric alcohols, such as cetyl alcohol (16 carbons) or myricyl alcohol (30 carbons), are the primary higher alcohols present in natural waxes.

Statement C is true: Gangliosides are complex acidic glycosphingolipids containing one or more N-acetylneuraminic acid (NANA/sialic acid) residues, found in highest concentrations in the ganglion cells and gray matter of the central nervous system.

Monosaccharides change from open chain to cyclic through:

- A. Hydroxylation
- B. Carbation
- C. Nucleophilic addition
- D. Hydrogenation

Correct Answer: C. Nucleophilic addition

Biochemical Rationale:

In aqueous solution, monosaccharides with five or more carbons exist predominantly in cyclic ring forms (>99%). Cyclization occurs via an intramolecular nucleophilic addition reaction, wherein an oxygen atom from a hydroxyl group elsewhere on the sugar molecule acts as a nucleophile and attacks the electrophilic carbonyl carbon (carbon 1 in aldoses, carbon 2 in ketoses), forming a cyclic hemiacetal or hemiketal ring.

The phospholipid important in the inner mitochondrial membrane (20% of total lipids):

- A. Cephalin
- B. Plasmalogen
- C. Lecithin
- D. Cardiolipin

Correct Answer: D. Cardiolipin

Biochemical Rationale:

Cardiolipin (diphosphatidylglycerol) consists of two phosphatidic acid molecules linked together by a central glycerol bridge. In eukaryotic cells, cardiolipin is localized almost exclusively to the inner mitochondrial membrane, where it constitutes approximately 20% of total membrane lipids. It is essential for stabilizing the tertiary/quaternary structure and function of electron transport chain respiratory complexes (Complexes I–IV).

Which GAG does not form proteoglycan?

- A. Heparin
- B. Keratan sulfate
- C. Hyaluronic acid
- D. Chondroitin sulfate

Correct Answer: C. Hyaluronic acid

Biochemical Rationale:

All major classes of glycosaminoglycans (GAGs)—including chondroitin sulfate, keratan sulfate, dermatan sulfate, heparin, and heparan sulfate—are synthesized in the Golgi and covalently attached via a trihexoside linkage to a core protein to form proteoglycan monomers.

Hyaluronic acid (hyaluronate) is the sole exception among GAGs: it is unsulfated, synthesized at the plasma membrane, and is never covalently attached to a core protein. Instead, proteoglycan monomers bind non-covalently to a central strand of hyaluronic acid via link proteins to construct giant proteoglycan aggregates in cartilage.

[24/09/2026 9:32 AM] Imaanaa Harmee: Cephalin consists of:

- A. Glycerol, fatty acid, phosphoric acid, choline
- B. Glycerol, fatty acid, phosphoric acid, ethanolamine
- C. Glycerol, fatty acid, phosphoric acid, inositol
- D. Glycerol, fatty acid, phosphoric acid, serine

Correct Answer: B. Glycerol, fatty acid, phosphoric acid, ethanolamine

Biochemical Rationale:

Cephalin is the common historical name for phosphatidylethanolamine, a major membrane glycerophospholipid. It consists of a glycerol backbone esterified to two long-chain fatty acids at C-1 and C-2, with a phosphate group attached at C-3 bound to the nitrogenous base ethanolamine.

Comparison: Phosphatidylcholine (lecithin) contains choline; phosphatidylinositol contains inositol; phosphatidylserine contains serine.

False statement:

- A. In mammal, major fat in adipose tissue is triglycerides
- B. Gangliosides derived from glucosylceramide contain more sialic acid molecules
- C. Glycosphingolipids are combination of ceramide and one or more sugar residues
- D. Glycosphingolipids are combination of sphingosine and galactose

Correct Answer: D. Glycosphingolipids are combination of sphingosine and galactose

Biochemical Rationale:

Why D is FALSE: Glycosphingolipids consist of ceramide (which is sphingosine linked to a long-chain fatty acid via an amide bond) covalently attached to one or more sugar residues. Sphingosine combined with galactose without the fatty acid is a psychosine derivative, not a complete glycosphingolipid.

Why A, B, and C are TRUE:

A: Triacylglycerols (triglycerides) represent the primary energy storage lipid in mammalian adipose tissue.

B: Gangliosides are complex acidic glycosphingolipids derived from glucosylceramide that contain one or more N-acetylneuraminic acid (NANA/sialic acid) residues.

C: By definition, glycosphingolipids are composed of a ceramide backbone plus carbohydrate head groups.

Incorrect about competitive inhibition:

A. Competes with substrates for active site

B. Increasing substrate concentration may reverse the reaction

C. No change in K_m

D. No change in V_{max}

Correct Answer: C. No change in K_m

Biochemical Rationale:

Why C is INCORRECT (The False Statement): In competitive inhibition, the inhibitor competes directly with the substrate for binding at the active site. This competition reduces the affinity of the enzyme for its substrate, which increases the apparent K_m . Thus, stating "No change in K_m " is false.

Why A, B, and D are correct properties: Increasing the substrate concentration can overcome and reverse competitive inhibition, and the maximal velocity (V_{max}) remains unchanged at saturating substrate levels.

An amino acid used to synthesize many hormones is:

A. Valine

B. Phenylalanine

C. Alanine

D. Histidine

Correct Answer: B. Phenylalanine

Biochemical Rationale:

Phenylalanine is an essential amino acid that is hydroxylated by phenylalanine hydroxylase to form Tyrosine. Tyrosine serves as the obligatory precursor for multiple crucial hormones and neurotransmitters:

Catecholamines: Dopamine, Norepinephrine, and Epinephrine (adrenal medullary hormone)

Thyroid Hormones: Triiodothyronine (T3) and Thyroxine (T4)

Melanin pigment

Correct order of organelles for export out of the cell:

- A. RER → SER → Golgi body
- B. SER → RER → Golgi body
- C. RER → Golgi body → SER
- D. Golgi body → RER → SER

Correct Answer: A. RER → SER → Golgi body

Biochemical Rationale:

Proteins destined for secretion or cell-surface export are synthesized by membrane-bound ribosomes on the Rough Endoplasmic Reticulum (RER). They are translocated into the lumen, pass through transition vesicles near the Smooth Endoplasmic Reticulum (SER) exit sites, and travel to the cis-face of the Golgi body for post-translational modification, sorting, and packaging into secretory vesicles prior to exocytosis.

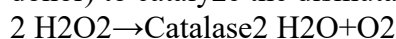
[24/09/2026 9:32 AM] Imaanaa Harmee: Enzyme which uses hydrogen acceptor as substrate:

- A. Xanthine oxidase
- B. Aldehyde oxidase
- C. Catalase
- D. Tryptophan oxygenase

Correct Answer: C. Catalase

Biochemical Rationale:

Catalase is a peroxisomal hemoprotein oxidoreductase that detoxifies reactive oxygen species. It uniquely utilizes hydrogen peroxide (H₂O₂) as a hydrogen acceptor substrate (as well as a donor) to catalyze the dismutation reaction:



[24/09/2026 9:32 AM] Imaanaa Harmee: Repeating units of hyaluronic acid are:

- A. N-acetyl glucosamine and D-glucuronic acid
- B. N-acetyl galactosamine and D-glucuronic acid
- C. N-acetyl glucosamine and galactose
- D. N-acetyl galactosamine and L-iduronic acid

Correct Answer: A. N-acetyl glucosamine and D-glucuronic acid

Biochemical Rationale:

Hyaluronic acid (hyaluronate) is an unbranched glycosaminoglycan (GAG) composed of repeating disaccharide units consisting of D-glucuronic acid (an acidic sugar) and N-acetylglucosamine (an N-acetylated amino sugar). Unlike most other GAGs, hyaluronic acid is unsulfated and does not covalently attach to a core protein.

Phospholipid is produced by:

- A. Nucleus
- B. Smooth endoplasmic reticulum
- C. Endoplasmic reticulum
- D. Cytoplasm

Correct Answer: B. Smooth endoplasmic reticulum

Biochemical Rationale:

Most cellular glycerophospholipids and triacylglycerols are synthesized by membrane-bound enzymes located in the smooth endoplasmic reticulum (SER). From the SER, newly synthesized phospholipids are transported via vesicles or lipid transfer proteins to the Golgi apparatus, organellar membranes, or the plasma membrane.

Heteropolysaccharide chondroitin differs from hyaluronic acid by presence of:

- A. N-acetyl glucosamine
- B. N-acetyl galactosamine
- C. D-glucuronic acid
- D. Galactose

Correct Answer: B. N-acetyl galactosamine

Biochemical Rationale:

Both chondroitin sulfate and hyaluronic acid contain D-glucuronic acid as their acidic sugar component. However, while hyaluronic acid contains N-acetylglucosamine (GlcNAc), chondroitin sulfate contains N-acetylgalactosamine (GalNAc, typically sulfated at C-4 or C-6) as its amino sugar.

Denaturation of proteins results in:

- A. Disruption of primary structure
- B. Breakdown of peptide bonds
- C. Destruction of hydrogen bonds
- D. Irreversible changes in the molecule

Correct Answer: C. Destruction of hydrogen bonds

Biochemical Rationale:

Protein denaturation involves the unfolding and disorganization of a protein's secondary and tertiary structures caused by agents (such as heat, extreme pH, or organic solvents) that disrupt non-covalent interactions—primarily hydrogen bonds, hydrophobic interactions, and ionic bonds.

Importantly, denaturation does not break peptide bonds (the primary covalent backbone remains intact). (Note: Some single-subunit denatured proteins can refold under ideal conditions, so denaturation is not universally irreversible).

Role of cholesterol in cell membrane:

- A. Acts regulating its fluidity
- B. Prevents formation of bumps in lipid bilayer
- C. Protects its integrity
- D. Participates in co-transport systems

Correct Answer: A. Acts regulating its fluidity

Biochemical Rationale:

Cholesterol is interspersed among membrane phospholipids and functions as a fluidity buffer. At elevated temperatures, its rigid steroid ring nucleus restricts excessive phospholipid movement; at lower temperatures, it fits into pockets created by cis-unsaturated fatty acids to prevent hydrophobic chains from packing too tightly, thereby maintaining optimal membrane fluidity.

Tertiary structure of protein describes:

- A. Order of amino acids
- B. Location of disulfide bonds
- C. Loop region of protein
- D. Ways of protein folding

Correct Answer: D. Ways of protein folding

Biochemical Rationale:

Tertiary structure defines the overall three-dimensional spatial conformation and full pattern of folding of the polypeptide chain. It is stabilized by non-covalent side-chain interactions (hydrophobic effect, hydrogen bonding, ionic interactions) as well as covalent disulfide linkages between distant cysteine residues.